

Blockchain Honor Minor Degree : Sem VIII

HBCC801 : Decentralized Finance (DeFi)

Module - 3: DeFi Primitives and Business Models

(9 Hours)

Instructors : Lifna C S

Topics to be covered

1. [DeFi Components](#): Blockchain, Cryptocurrency The Smart Contract, Platform, Oracles, Stablecoins, Decentralized Applications
2. [DeFi Primitives](#):
 - Transactions Fungible Token: Equity Tokens, Utility Tokens and Governance Tokens
 - NFT: NFT Standard, Multi-token standard
 - Custody Supply Adjustment: Burn-Reduce Supply, Mint-Increase Supply, Bonding Curve
 - Pricing Supply Incentives: Staking Rewards, Slashing, Direct Rewards and Keepers, Fees
 - Swap: Order Book Matching, Automated Market Makers, Collateralized Loans Flash Loans (Uncollateralized Loans)
3. [DeFi Key Metrics](#): Total Value Locked, Daily Active Users, Market Cap
4. [DeFi Major Business Models](#): Decentralized Currencies, Decentralized Payment Services, Decentralized fundraising, Decentralized Contracting.

1. DeFi Components:

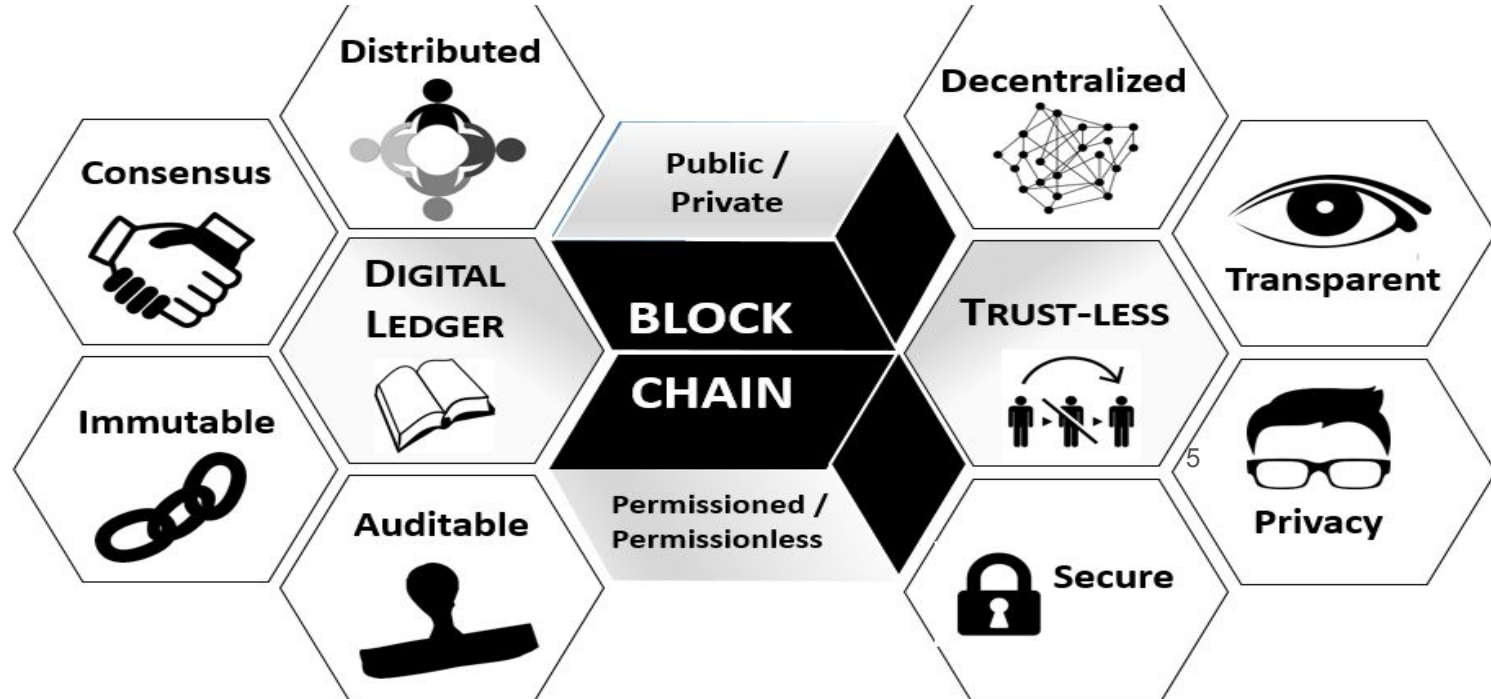
1. [Blockchain](#)
2. [Cryptocurrency](#)
3. [Smart Contract,](#)
4. [Platforms](#)
5. [Oracles](#)
6. [Stablecoins](#)
7. [Decentralized Applications \(DApps\)](#)

1. DeFi Components - Blockchain

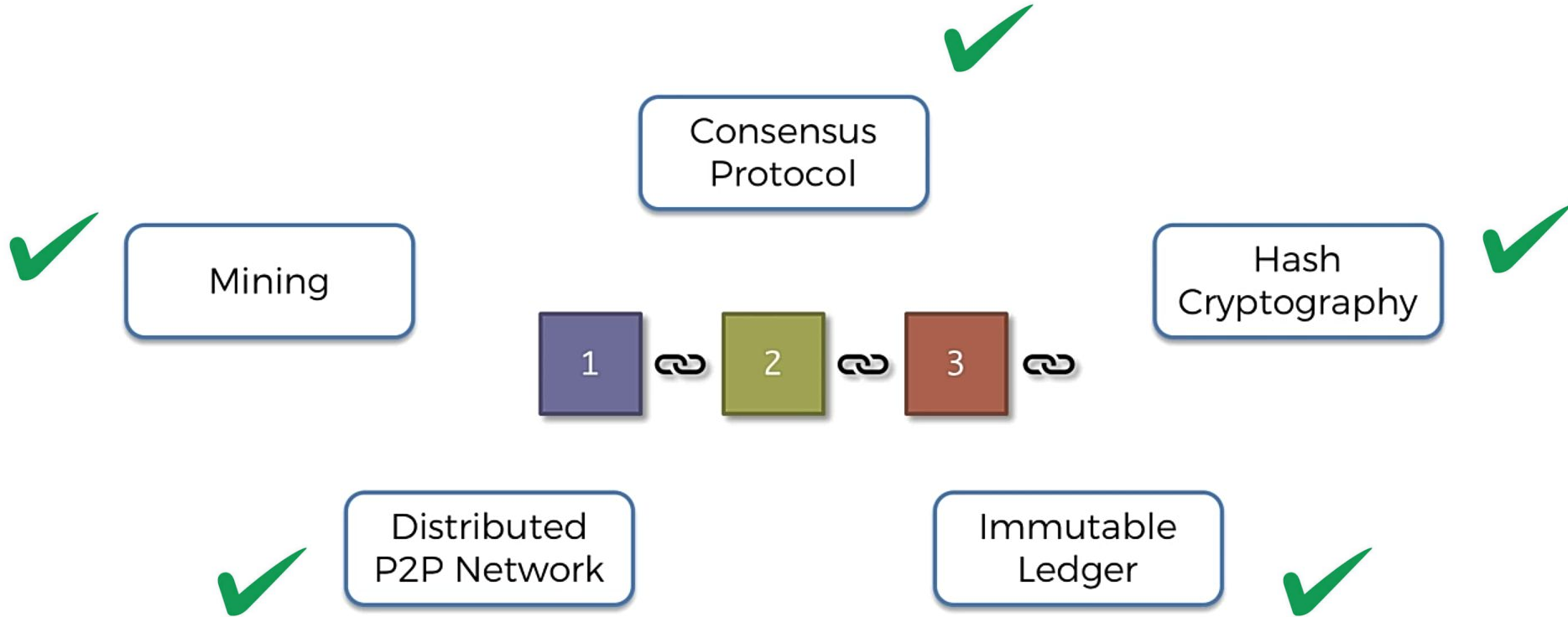
- ✓ Blockchain is a **digital public ledger** that records **online transactions**.
- ✓ Blockchain ensures **confidentiality, integrity and privacy**.
- ✓ When a **new block** is added to a blockchain, it is **linked to the previous block** using a **cryptographic hash**.
- ✓ This ensures the **chain is never broken** and that **each block is permanently recorded**.
- ✓ It is also **intentionally difficult** to alter past transactions in blockchain since **all the subsequent blocks must be altered first**.

1. DeFi Components - Blockchain

Characteristics of Blockchain



1. DeFi Components - Blockchain



1. DeFi Components - Cryptocurrency

Cryptocurrency as a **Form of Currency**

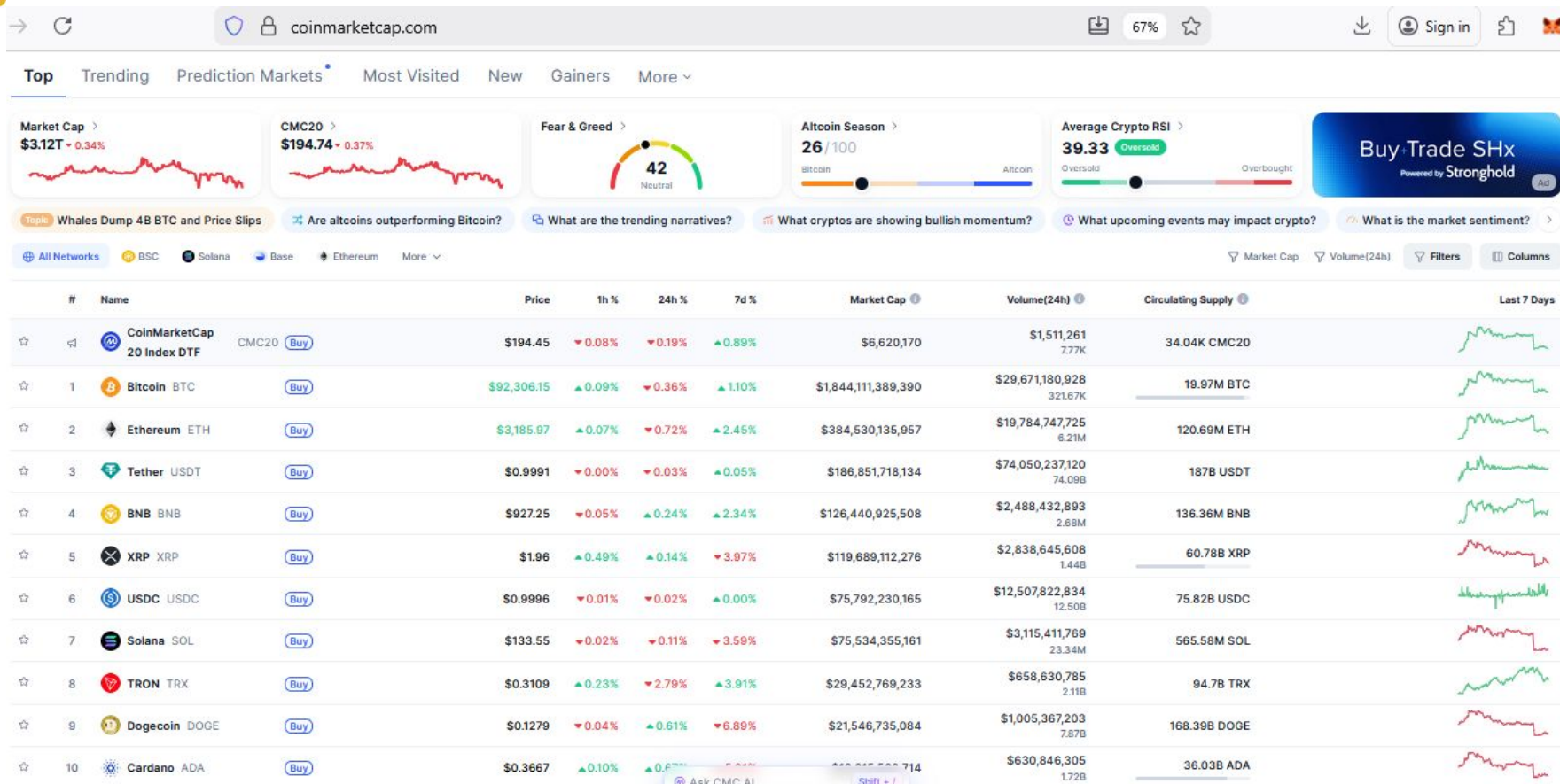
- **Cryptos** - Eg. **Bitcoin, Litecoin, Shiba Inu, Dogecoin** etc...
- Various companies and even countries around the world accept some of these digital currencies for conducting transactions.
- However, the **high volatility** of **Bitcoin** and other popular cryptocurrencies makes it unsuitable for everyday use by the public.

1. DeFi Components - Cryptocurrency

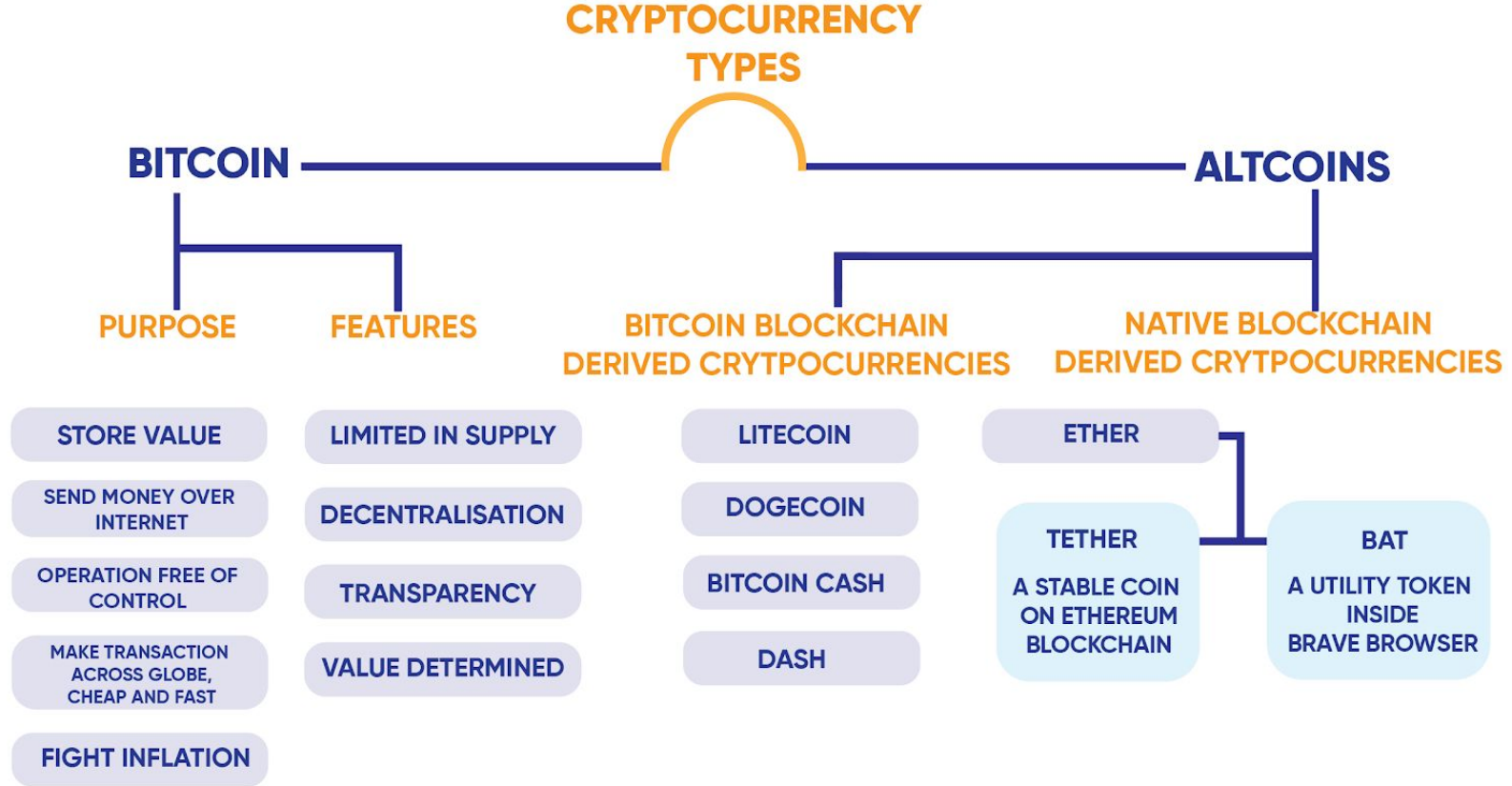
List of Latest Cryptocurrencies to Invest in 2025!

RANK	APPS		RANK	APPS	
1		Bitcoin	11		Litecoin
2		Ethereum	12		Chainlink
3		Ripple	13		Tron
4		Solana	14		Tether
5		Dogecoin	15		Dai
6		Binance Coin	16		SKALE
7		Cardano	17		PEPE
8		Avalanche	18		Uniswap
9		Polkadot	19		Bitcoin Cash
10		Stellar	20		Cosmos

1. DeFi Components - Cryptocurrency



1. DeFi Components - Cryptocurrency



1. DeFi Components - Cryptocurrency

BitCoin

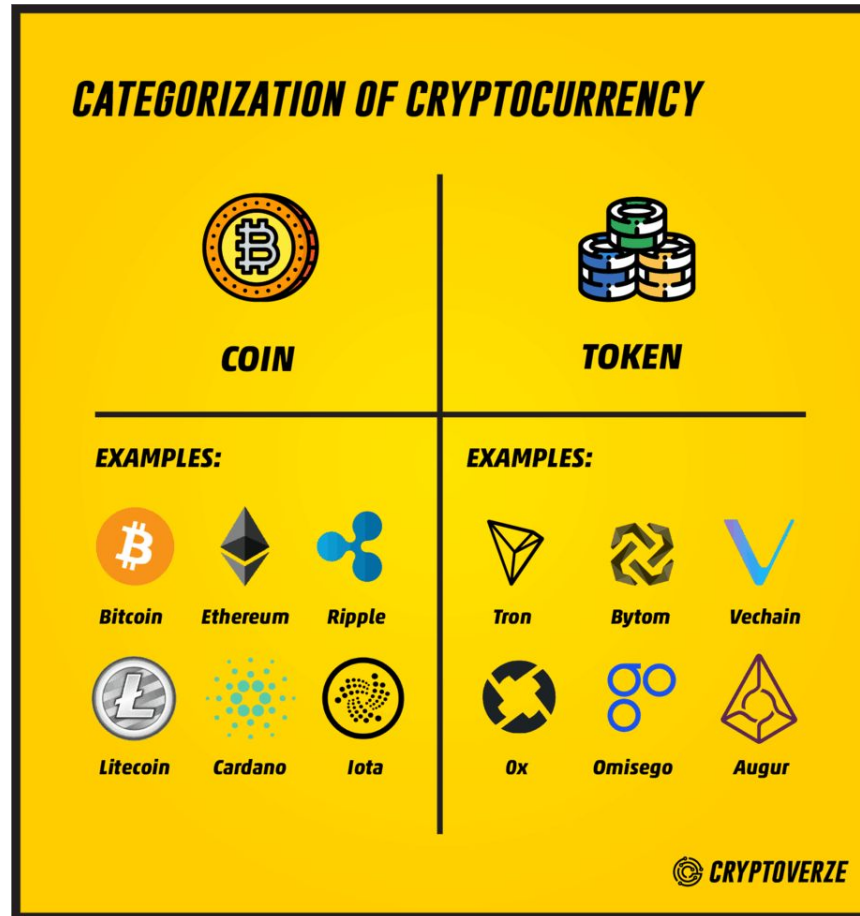
- Launched in **2009**
- **world's largest cryptocurrency** by market capitalization.
- Bitcoin is **created, distributed, traded, and stored using a decentralized ledger system (Blockchain)**
- Bitcoin and its ledger are **secured by proof-of-work (PoW) consensus**,
- Bitcoin can be **purchased via various cryptocurrency exchanges**.
- **first decentralized virtual currency**
- provides **secure global transactions quickly and without third-party manipulations**.
- **created to address the inefficiencies in global financial systems**.
- Bitcoin is **not issued by any government**, and **banks do not manage accounts or validate transactions**.
- It is **based on a cryptographic system** that uses certain codes and numbers to keep information safe and secure.
- **stored in digital wallets that allow users to manage and trade their coins**.
- **currently accepted as a means of payment for products sold or services provided**.
- **offers lower transaction fees** compared to traditional online payment mechanisms.
- However, **the value of Bitcoin has been extremely volatile over the past few years**.

1. DeFi Components - Cryptocurrency

Altcoins or Alternative Coins

- Describe [all cryptocurrencies other than Bitcoin](#).
- These coins also **use blockchain technology** that allows secure peer-to-peer transactions.
- Altcoins were built on the success of Bitcoin by **slightly changing the rules to appeal to different types of users**.
- They essentially **solve the inefficiencies of Bitcoin**.
 - **Litecoin** - address issues related to scalability, higher transaction time & charges, and environmental concerns
 - **Ether** - created based on the idea that blockchain tech can be used to create applications that go beyond just enabling a digital currency.
- There are **more than 10,000 altcoins** in existence today.
- Altcoins come in **several types based on what they were designed for**.

1. DeFi Components - Cryptocurrency

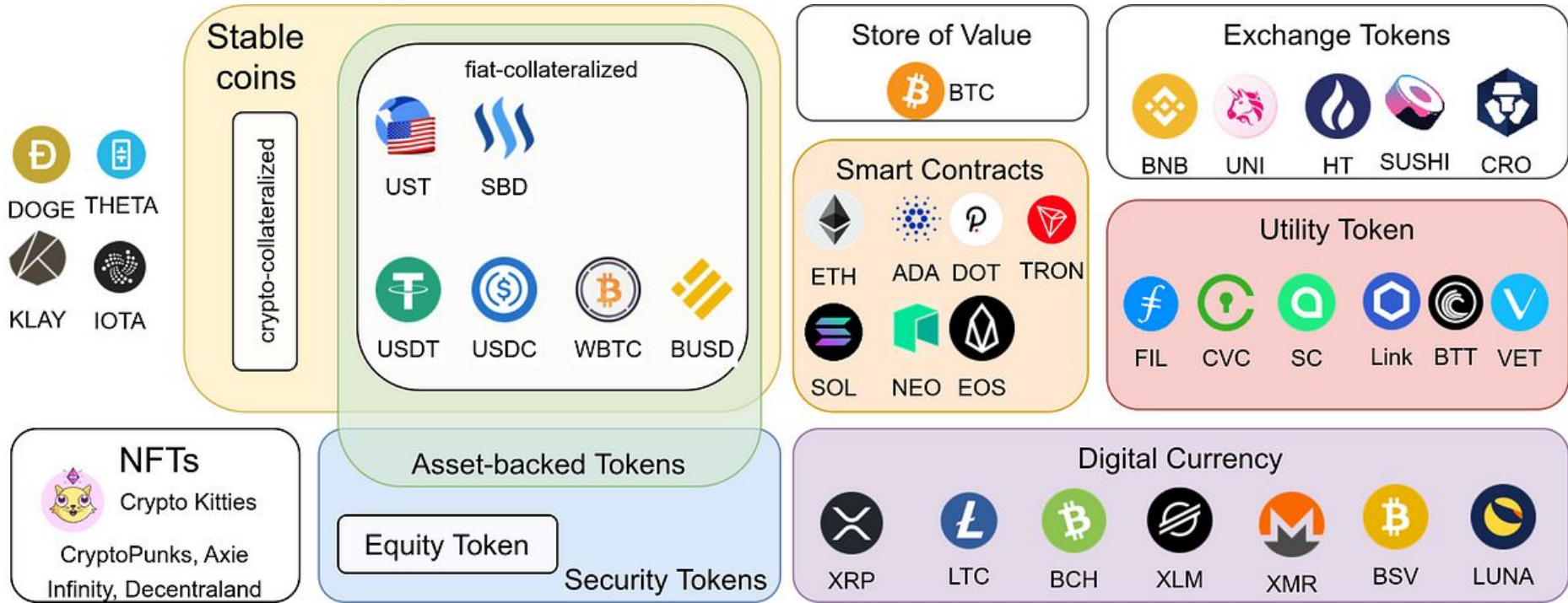


Courtesy : [CoinCrunch - Medium](https://www.coincrunch.net/)

1. DeFi Components - Cryptocurrency



1. DeFi Components - Cryptocurrency

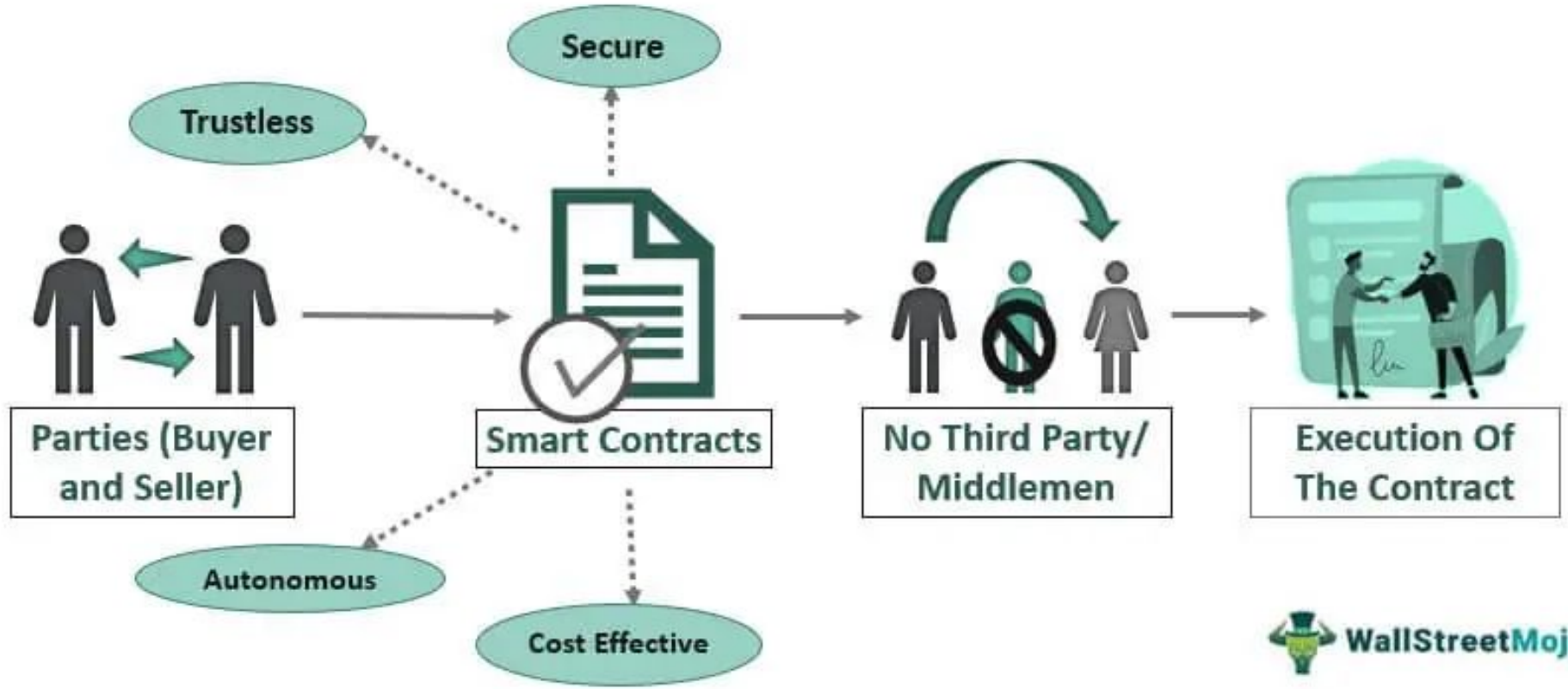


1. DeFi Components - Smart Contract

- A **self-executing program** that automates the actions required in an agreement or contract. Once completed, the transactions are trackable and irreversible.
- Smart contracts **permit trusted transactions and agreements to be carried out among disparate, anonymous parties** without the need for a central authority, legal system, or external enforcement mechanism.
- **do not contain legal language**, terms, or agreements
- **Only code that executes actions when specified conditions** are met.
- Nick Szabo, an American computer scientist
 - invented a virtual currency called "**Bit Gold**" in 1998,
 - defined smart contracts as computerized transaction protocols that execute the terms of a contract



1. DeFi Components - Smart Contract



1. DeFi Components - Smart Contract Benefits



Autonomy



Accuracy



Transparency



High Speed



Data storage



Trustability



Cost Savings



Robust Backup

1. DeFi Components - Blockchain Platforms Comparison

	Public	Private	Hybrid	Consortium
 Permissioned/Permissionless	Permissionless	Permissioned	Permissioned & Permissionless	Permissioned
 Control	No control by a central authority	Control by a central authority	Control by a central authority	Control by multiple central authorities
 Main Advantages	✓ Independence ✓ Transparency	✓ Performance ✓ Scalability	✓ Performance ✓ Low Cost	✓ Performance ✓ Security
 Main Disadvantages	✗ Performance ✗ Scalability Issues	✗ Security ✗ Trust	✗ Transparency ✗ Upgrading	✗ Transparency
 Examples	Bitcoin Litecoin	Hyperledger Fabric	XRP token	Corda Quorum

1. DeFi Components - Top 5 Blockchain Platforms for 2025

1. **Ethereum** : popular for decentralized applications (dApps) and smart contracts,
Layer 2 scaling solutions.
2. **Solana**: exceptional transaction speed and low fees
(best for DeFi, NFTs, and high-performance applications).
3. **Hyperledger Fabric** :
 - popular permissioned, modular blockchain platform
 - designed for enterprise-grade applications (privacy and security requirements)
4. **Polkadot** : focus on **interoperability** between blockchains.
Uses parachains connected to a central relay chain.
blockchains that can securely communicate with each other.
5. **Binance Smart Chain (BNB Chain)** :
 - EVM-compatible and optimized for **fast, low-cost transactions**.
 - Strong DeFi adoption and easy migration from Ethereum.
 - Widely used for DApps and token ecosystems

1. DeFi Components - Comparison of Top Blockchain Platforms

Feature	Ethereum	Hyperledger Fabric	Solana	Polkadot	BNB Chain
Type	Public	Permissioned	Public	Public	Public
Primary Use	DeFi, NFTs, DAOs, DApps	Enterprise & Govt solutions	High-speed DeFi, NFTs, Gaming	Interoperable blockchains	Low-cost DeFi & DApps
Smart Contracts	Yes (Solidity, Vyper)	Yes (Chaincode)	Yes (Rust, C)	Yes (Substrate-based)	Yes (Solidity – EVM)
Consensus	Proof of Stake (PoS)	Raft / PBFT	PoH + PoS	Nominated PoS	Proof of Staked Authority
Permission Model	Permissionless	Permissioned	Permissionless	Permissionless	Permissionless
Transaction Speed	Moderate (Layer-2 improves)	High	Very High	High	High

1. DeFi Components - Comparison of Top Blockchain Platforms

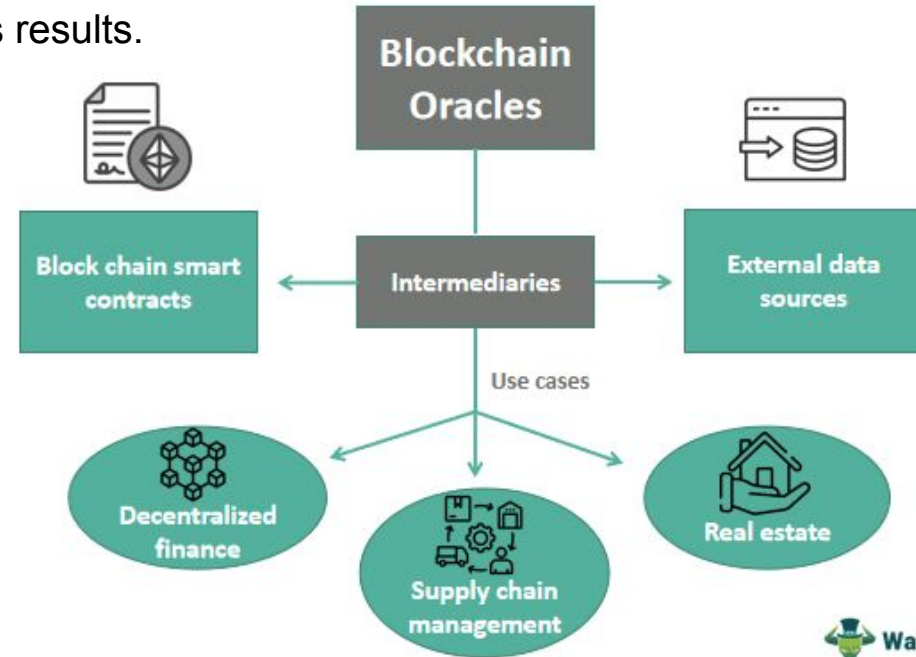
Feature	Ethereum	Hyperledger Fabric	Solana	Polkadot	BNB Chain
Transaction Cost	High / Variable	Low	Very Low	Low	Very Low
Decentralization	High	Low–Medium	Medium	High	Medium
Scalability	Via Layer-2s	Native	Native	Parachains	Native
Token Needed	ETH	No native token	SOL	DOT	BNB
Enterprise Adoption	Medium	Very High	Low–Medium	Medium	Medium
Governance	Community driven	Organization controlled	Foundation-led	On-chain governance	Binance influenced

1. DeFi Components - Oracles

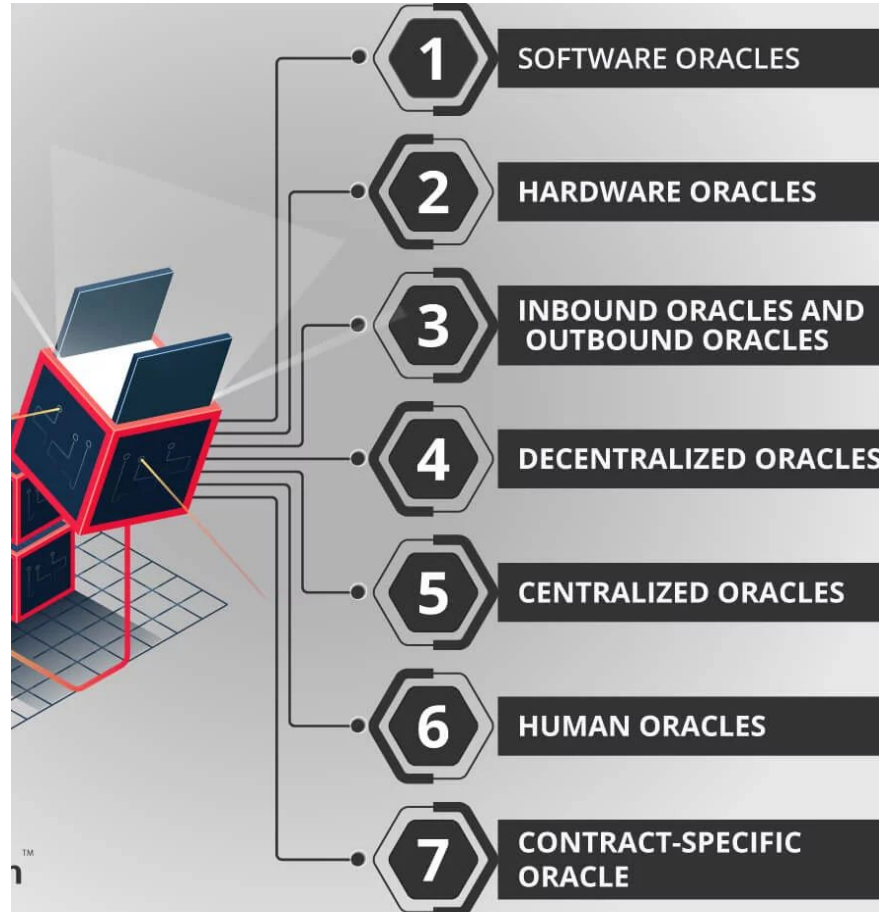
- A bridge between blockchain and the external world.
- It supplies smart contracts with real-world data.
- Example: price of ETH/USD, weather data, sports results.

Need for Oracles

- Smart contracts cannot access off-chain data by themselves.
- Oracles enable real-world use cases such as:
 - DeFi price feeds
 - Insurance payouts
 - Prediction markets



1. DeFi Components - Types of Oracles



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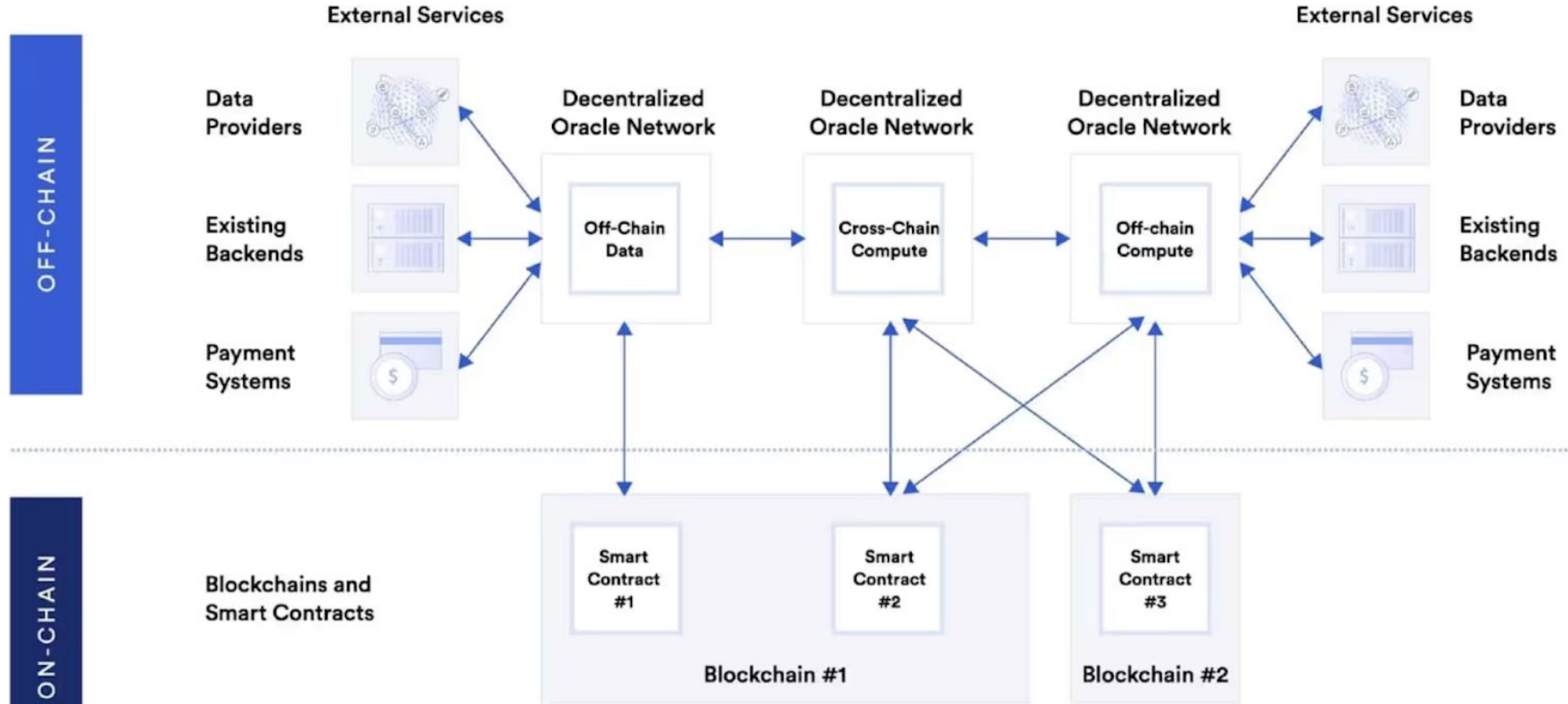
1. DeFi Components - Types of Oracles

Oracle Type	Data Source	Direction	Centralization	What It Does	Example Use Case	Key Risk
Software Oracle	Online APIs, websites, databases	Inbound	Can be centralized or decentralized	Brings digital data to blockchain	Crypto price feeds, stock prices	API manipulation
Hardware Oracle	IoT devices, sensors	Inbound	Usually centralized	Brings physical-world data	Temperature for crop insurance	Sensor tampering
Inbound Oracle	Off-chain → On-chain	Inbound	Either	Sends real-world data to smart contracts	ETH/USD price	Data integrity
Outbound Oracle	On-chain → Off-chain	Outbound	Either	Sends blockchain data to real systems	Trigger bank payment	Execution failure
Cross-chain Oracle	Other blockchains	Both	Usually decentralized	Transfers data between blockchains	Bridge price feeds	Bridge exploits

1. DeFi Components - Types of Oracles

Oracle Type	Data Source	Direction	Centralization	What It Does	Example Use Case	Key Risk
Centralized Oracle	Single data provider	Both	Centralized	One source provides data	Simple dApps	Single point of failure
Decentralized Oracle	Multiple data providers	Inbound	Decentralized	Aggregates data from many sources	DeFi price feeds	Higher cost
Consensus-based Oracle	Multiple nodes + voting	Inbound	Decentralized	Uses consensus to verify data	Prediction markets	Collusion risk
Human Oracle	Individuals / experts	Inbound	Semi-centralized	Humans verify and submit data	Legal outcomes, sports results	Bias, error

1. DeFi Components - Types of Oracles

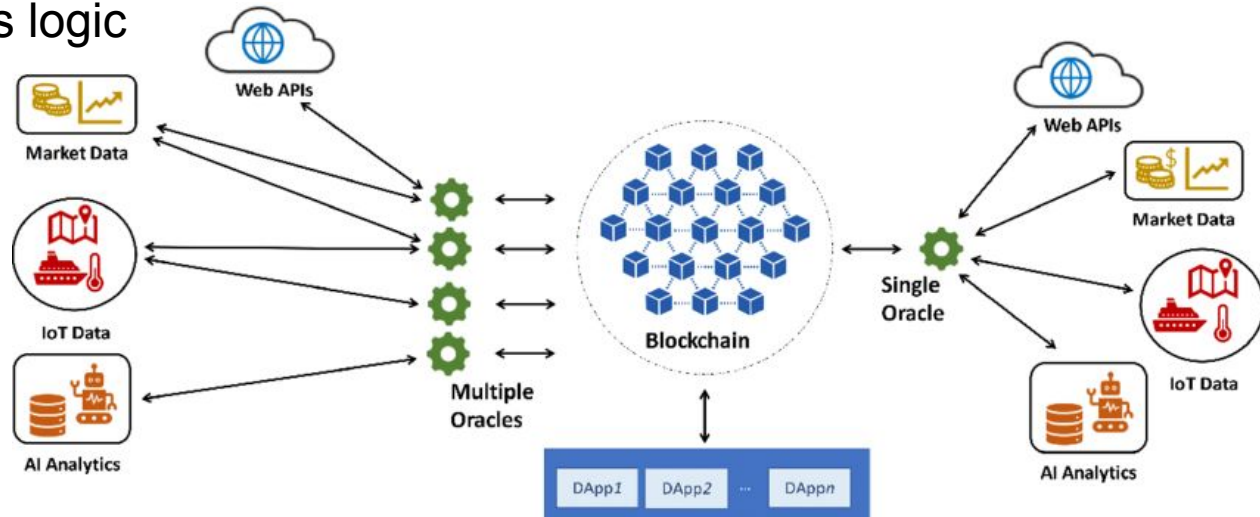


1. DeFi Components - Industry based Oracle Use Cases

Industry	Oracle Data Used	Example Platform	Outcome
DeFi	Asset prices	Chainlink, Aave	Accurate collateral valuation
Insurance	Flight, weather data	Etherisc, Arbol	Automated claim payouts
Supply Chain	IoT sensor data	VeChain	Real-time product tracking
Gaming	Random numbers, game state	Chainlink VRF	Provably fair gameplay
Healthcare	Patient, drug records	Solve Care, MediLedger	Fraud prevention, authenticity
Energy	Production, consumption	LO3 Energy	Peer-to-peer energy trading
Betting	Sports results	Augur, Chiliz	Automated bet resolution

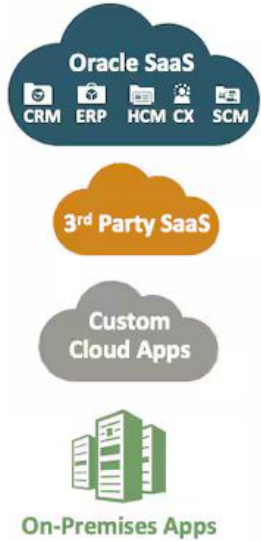
1. DeFi Components - Working of Oracle

1. Smart contract requests data
2. Oracle fetches data from external source
3. Oracle validates data
4. Oracle sends data on-chain
5. Smart contract executes logic



1. DeFi Components - Working of Oracle

Enterprise Systems and Front-end Apps



Integration

Adapters

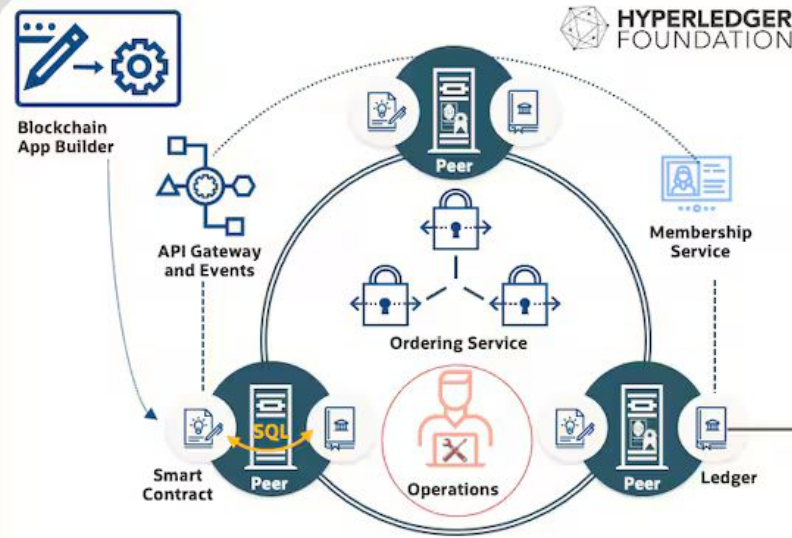
REST APIs

Reliable Events

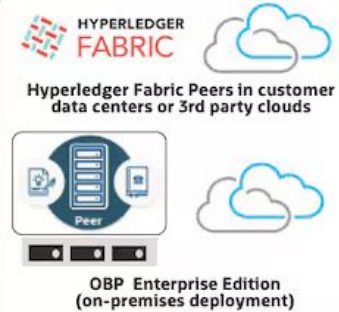
Hyperledger

Client SDKs

OCI Blockchain Platform



Other Nodes

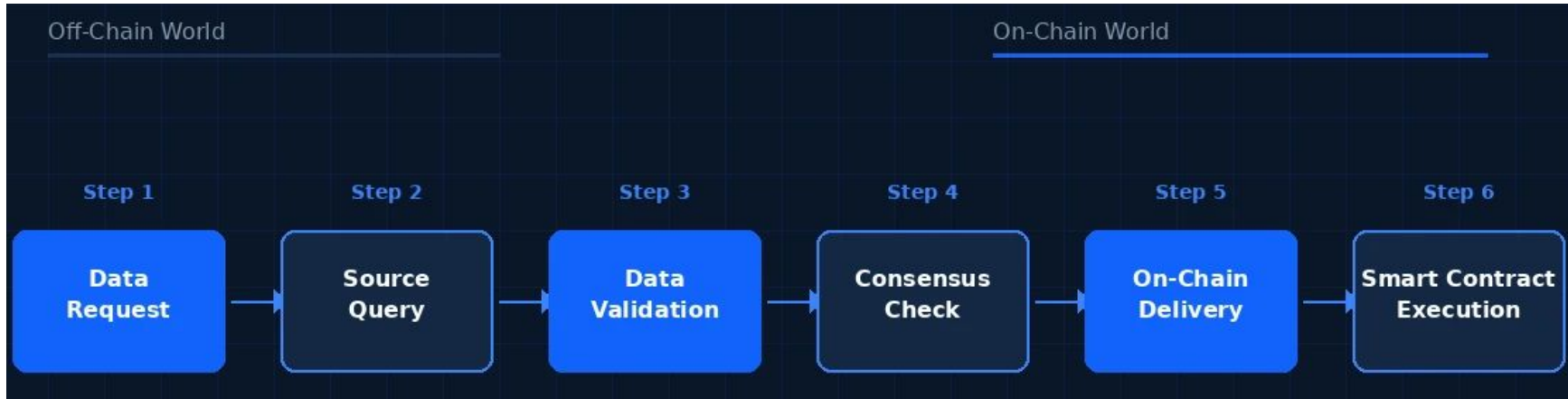


Off-Chain Storage and Live Analytics



1. DeFi Components - Oracle Life Cycle

1. Data Request
2. Data Collection
3. Data Verification
4. Data Transmission
5. Smart Contract Execution



1. DeFi Components - Oracle

Oracle Risks

- Data manipulation
- Single point of failure
- Delayed updates
- Smart contract dependency

Popular Oracle Networks

- Chainlink
- Band Protocol
- API3
- UMA

1. DeFi Components - Stablecoins

- Class of cryptocurrencies backed by reserve assets (cash or commodity).
- 
- **offer price stability** while ensuring all basic features or benefits of cryptocurrencies.
 - Provides **instant processing and security of payments**.
 - **1 Tether (USDT)** is pegged to (or **backed by**) **1 US Dollar**, as fiat currencies are pegged to an underlying asset such as gold or foreign exchange reserves, **their valuations remain free from wild movements**.
 - **PAX Gold**, a digital token **backed by physical gold**. (currently, **1 PAXG is nearly 30 grams**)
 - The basket is meant to act as a reserve to redeem holders if the cryptocurrency fails or faces problems. Price fluctuations for stablecoins are not meant to exceed a narrow range.

1. DeFi Components - Comparison of Top 5 Stablecoins

Feature	USDT	USDC	BUSD	DAI	TUSD
Peg Type	1:1 USD	1:1 USD	1:1 USD	1:1 USD	1:1 USD
Backing / Reserve	Mixed assets (cash, loans, securities)	Fully backed by USD reserves	USD in regulated banks	Crypto-collateral (ETH, WBTC, etc.)	Fiat in escrow accounts
Issuer	Tether Ltd.	Circle & Coinbase	Paxos / Binance	MakerDAO	TrustToken
Decentralization	Centralized	Semi-centralized	Centralized	Decentralized	Semi-centralized
Main Use Cases	Trading, liquidity	Payments, lending	Trading, lending	Lending, savings	Trading, collateral
Transparency	Medium	High	High	On-chain collateral	High
Major Risks	Reserve trust, regulation	Custodian & regulation	Regulatory limits	Liquidation, oracle risk	Custodial risk

1. DeFi Components - Comparison of Types of Stablecoins

Aspect	Fully Collateralized Stablecoins	Algorithmic Stablecoins
Operational Model	Backed by tangible assets in secure escrow.	Uses algorithms to dynamically adjust supply.
Basis of Value	Value tied to reserve of tangible assets.	Value maintained through algorithmic adjustments.
Redemption Guarantee	Redeemable for equivalent in reserve asset.	Redemption tied to algorithmic stabilization, not specific asset.
Price Stability Mechanism	Stability from backing tangible assets.	Stability maintained by adjusting token supply based on demand.
Market Flexibility	Less flexible due to direct link to collateral.	More adaptable, with supply adjustments responding to demand.

1. DeFi Components - DApps

- A software application that operates on a decentralized network, typically using blockchain technology.
- Compared to traditional applications that rely on centralized servers, Dapps leverage the security, transparency, and trustlessness of blockchain to function.
- Salient Features of DApps
 1. **Decentralization:**
 - Dapps run on a decentralized network of computers, often referred to as a blockchain.
 - This network is distributed across many nodes, making it resistant to single points of failure and censorship.
 2. **Smart Contracts:**
 - Dapps typically use smart contracts, which are self-executing contracts with the terms of the agreement directly written into code.
 - Smart contracts automatically execute actions when specific conditions are met, without the need for intermediaries.
 3. **Transparency:**
 - All transactions and actions within a Dapp are recorded on a public ledger (the blockchain).
 - This transparency ensures that anyone can verify transactions and data, enhancing trust in the application.



1. DeFi Components - DApps

4. Security:

- Blockchain technology, with its cryptographic techniques, provides a high level of security.
- Transactions are immutable, meaning once recorded on the blockchain, they cannot be altered.
- This makes Dapps resilient to fraud and hacking.

5. Trustlessness:

- Dapps aim to operate without relying on a central authority or intermediary.
- Users can interact with the application and each other directly without needing to trust a third party.

6. Token Economy:

- Many Dapps have their own native tokens or use established cryptocurrencies like Ethereum's Ether.
- These tokens can be used for various purposes within the application, such as paying for services, participating in governance, or earning rewards.

7. Use Cases:

- Dapps have a wide range of use cases, including decentralized finance (DeFi), non-fungible tokens (NFTs), supply chain management, voting systems, gaming, and more.
- Each Dapp is designed to solve specific problems or provide new functionalities in a decentralized manner.



1. DeFi Components - DApps

8. User Interface:

- Dapps often have user interfaces (UIs) similar to traditional apps or websites, making them accessible to a broader user base.
- Users may not even be aware that they are interacting with blockchain technology.

9. Challenges:

- While Dapps offer numerous advantages, they also face challenges, including scalability issues, user adoption barriers, and regulatory compliance.
- These challenges are actively being addressed by the blockchain community.

10. Development:

- Building a Dapp typically involves programming smart contracts using languages like Solidity (for Ethereum) and developing a frontend interface.
- Popular Dapp development frameworks like Truffle and web3.js make the development process more accessible.

1. DeFi Components - DApps Architecture

1. Ethereum blockchain:

- Securely executes and verifies application code, called smart contracts.
- DApps use the Ethereum blockchain for data storage.

2. Smart Contracts:

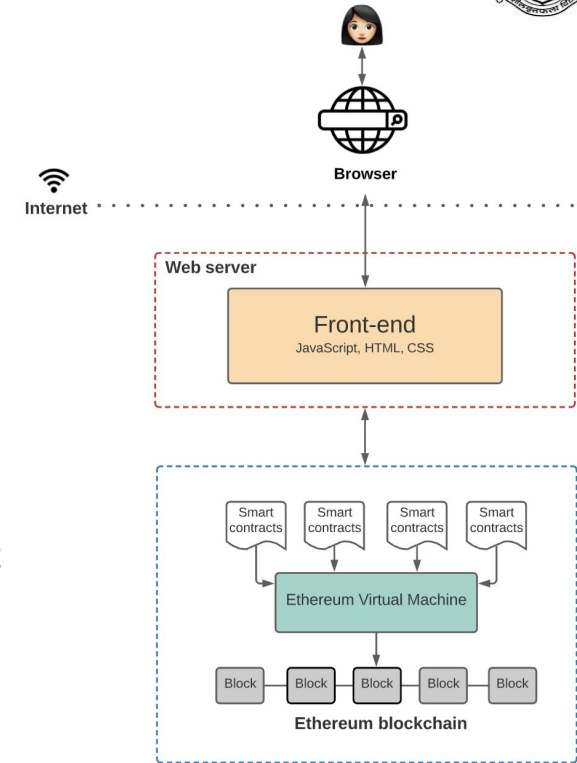
- DApps use smart contracts to define the state changes happening on the blockchain.
- A smart contract is a collection of code and data that resides at a specific address on the Ethereum Blockchain and runs on the Ethereum blockchain.

3. Ethereum Virtual Machine(EVM):

- Global virtual computer that executes the logic defined in the smart contracts and processes the state changes that happen on this Ethereum network.

4. Front-end:

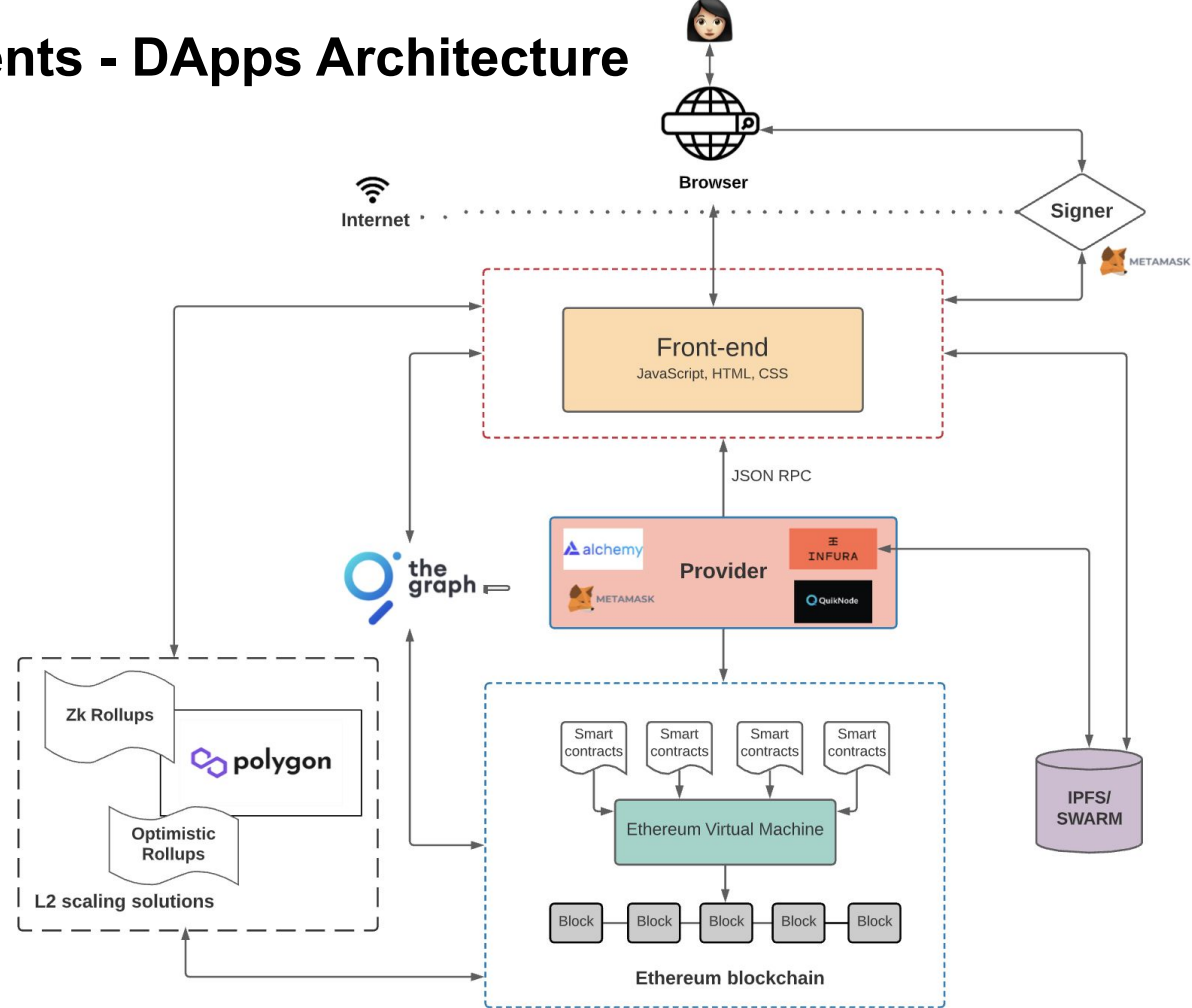
- The part of the DApps, that users can see and interact with such as the graphical user interface(GUI),
- Communicates with the application logic defined in smart contracts.



1. DeFi Components - DApps Architecture

Addressing

- Storage,
- Querying
- Scalability



2. DeFi Primitives:

- [Transactions](#)
- [Fungible Token](#): Equity Tokens, Utility Tokens and Governance Tokens
- [NFT](#): NFT Standard, Multi-token standard
- [Custody](#)
- [Supply Adjustment](#): Burn-Reduce Supply, Mint-Increase Supply, Bonding Curve, Pricing Supply
- [Incentives](#): Staking Rewards, Slashing, Direct Rewards and Keepers, Fees
- [Swap](#): Centralized Exchange, Decentralized Exchange, Order Book Matching, Automated Market Makers,
- Collateralized Loans
- Flash Loans (Uncollateralized Loans)

2. DeFi Primitives - Transaction Mechanism (Ethereum)

How do transactions work?

- Transactions involve sending data and/or ETH (or other tokens) from one address to another.
- An Ethereum user can control addresses through an externally owned account (EOA) or by using smart contract code (contract account).
- When sending data to a contract account, the data are used to execute code in that contract. The transaction may or may not have an accompanying ETH payment for use by the contract.
- Transactions sent to an EOA can only transfer ETH.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

How do transactions work?

- A single transaction starts with an end-user from an EOA, but can interact with a large number of dApps (or any Ethereum smart contract) before completing.
- The transaction starts by interacting with a single contract, which will enumerate all of the intermediate steps in the transaction required within the contract body.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

Atomicity

- Clauses in a smart contract can cause a transaction to fail and thereby revert all previous steps of the transaction; as a result, transactions are *atomic*.
- Atomicity is a critical feature of transactions because funds can move between many contracts (i.e., “exchange hands”) with the knowledge and security that if one of the conditions is not met, the contract terms reset as if the money never left the starting point.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

Gas

- Transactions have a gas fee, which varies based on the complexity of the transaction. E.g., low gas fee is used to compensate a miner for including and executing a transaction, and high gas fee for more data-intensive transactions
- If a transaction reverts for any reason, or runs out of gas, the miner forfeits all gas used until that point. Forfeiture protects the miners who, without this provision, could fall prey to large volumes of failed transactions for which they would not receive payment.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

Gas fees

- The gas price is determined by the market and effectively creates an auction for inclusion in the next Ethereum block.
- Higher gas fees signal higher demand and therefore generally receive higher priority for inclusion.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

Mempool

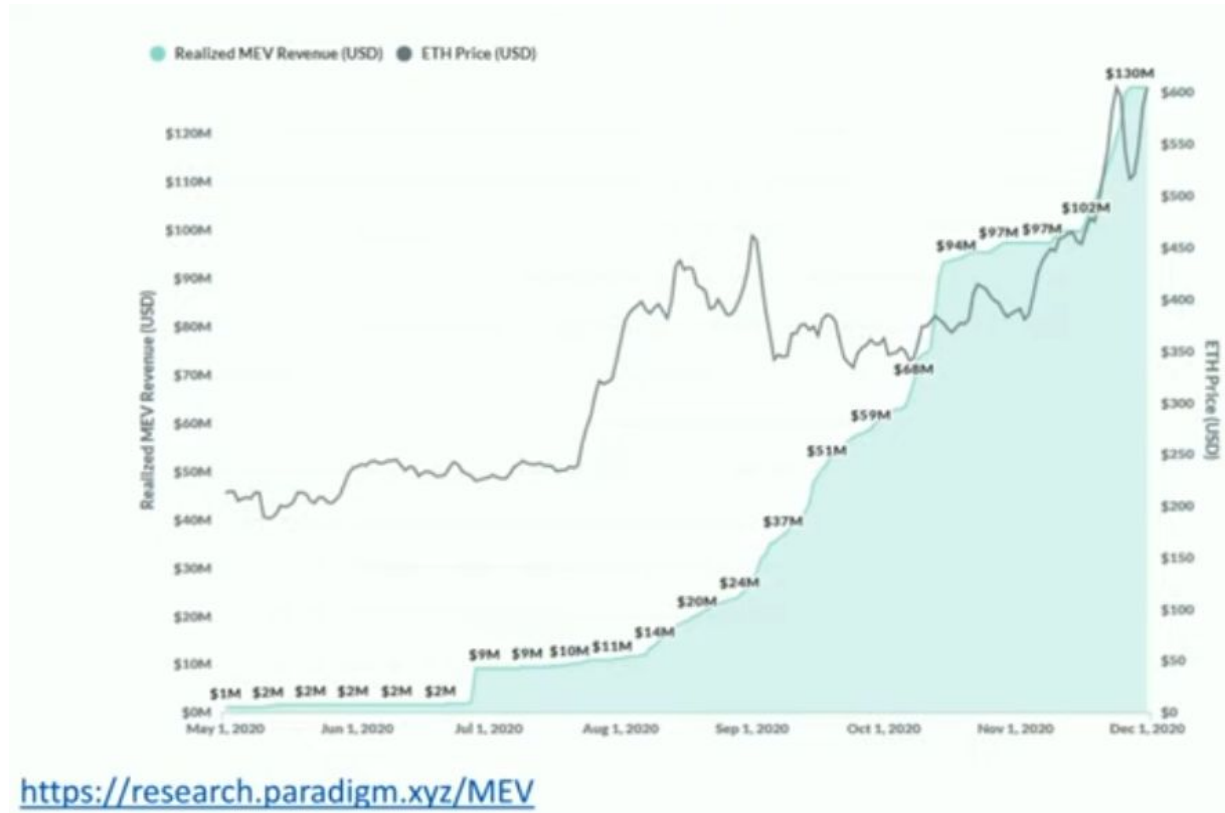
- Transactions are posted to a *memory pool*, or *mempool*, before they are added to a block.
- Miners monitor these posted transactions, add them to their own mempool, and share the transaction with other miners to be included in the next available block.
- If the gas price offered by the transaction is uncompetitive relative to other transactions in the mempool, the transaction is deferred to a future block.

2. DeFi Primitives - Transaction Mechanism (Ethereum)

Miner extractable value

- Any actor can see transactions in the mempool by running or communicating with mining nodes.
- This visibility can even allow for advanced “front-running”. This is not to be confused with the illegal front-running in centralized finance. If a miner sees a transaction in the mempool (and all transactions are public information), she could profit from by either executing herself or front-running it, the miner is incentivized to do so if lucky enough to win the block.

2. DeFi Primitives - Transaction Mechanism (Ethereum)



2. DeFi Primitives - Fungible Tokens

ERC-20 functionality

- When a token implements the ERC-20 interface, any application that generically handles the defined functionality can instantly and seamlessly integrate with the token.
- Using ERC-20 and similar interfaces, application developers can confidently support tokens that do not yet exist.

2. DeFi Primitives - Fungible Tokens

ERC-20 interface

- **totalSupply()**—read the token's total supply;
- **balanceOf(address)**—read the balance of the token for a particular user;
- **transferFrom(from address, to address, amount)**—send “amount” tokens from the balance of tokens held at “from address” to “to address”; and
- **approve(owner, spender, amount)**—allows “spender” to spend “amount” tokens of “owner” on behalf of owner.

2. DeFi Primitives - Fungible Tokens

Equity tokens

- An equity token (not traditional stocks) is simply a token that represents ownership of an underlying asset or pool of assets.
- The units must be fungible so that each corresponds to an identical share in the pool. For example, suppose a token, TKN, has a total fixed supply of 10,000, and TKN corresponds to an ETH pool of 100 ETH held in a smart contract.
- The smart contract stipulates that for every unit of TKN it receives, it will return a pro rata amount of ETH, fixing the exchange ratio at 100 TKN/1 ETH.

2. DeFi Primitives - Fungible Tokens

Equity tokens

- We can extend the example so the pool has a variable amount of ETH. Suppose the ETH in the pool increases at 5% per year by some other mechanism.
- Now 100 TKN would represent 1 ETH plus a 5% perpetual cash flow of ETH. The market can use this information to accurately price the value of TKN.

2. DeFi Primitives - Fungible Tokens

Equity tokens

- In actual equity tokens, the pools of assets can contain much more complex mechanics.
 - Variable interest-rate mechanics (Compound)
 - Contract that owns a multi-asset pool with a complex fee structure (Uniswap).
 - A standard interface for creating equity tokens with static or dynamic holdings (Set Protocol).

2. DeFi Primitives - Fungible Tokens

Utility tokens

- Utility tokens are fungible tokens that are required to utilize some functionality of a smart contract system or that have an intrinsic value proposition defined by its respective smart contract system.
- Examples of use cases for utility tokens:
 - To be collateral (e.g., SNX)
 - To represent reputation or stake (e.g., REP, LINK)
 - To maintain stable value relative to underlying or peg (e.g., DAI, Synthetix Synth)
 - To pay application-specific fees (e.g., ZRX, DAI, LINK)

2. DeFi Primitives - Fungible Tokens

Utility tokens

- The last example includes all stablecoins, regardless of whether the stablecoin is fiat collateralized, crypto-collateralized, or algorithmic.
- In the case of USDC, a fiat-collateralized stablecoin, the utility token operates as its own system without any additional smart-contract infrastructure to support its value.
- The value of USDC arises from the promise of redemption for USD by its backing companies, including Coinbase.

2. DeFi Primitives - Fungible Tokens

Governance tokens

- Governance tokens are similar to equity tokens in the sense they represent percentage ownership. Instead of asset ownership, governance token ownership applies to voting rights
- Many smart contracts have embedded clauses stipulating how the system can change; for instance, allowed changes could include adjusting parameters, adding new components, or even altering the functionality of existing components.

2. DeFi Primitives - Fungible Tokens

Governance tokens

- Any platform with admin-controlled functionality is not truly DeFi because of the admins' centralized control.
- A contract without the capacity for change is necessarily rigid, however, and has no way to adapt to bugs in the code or changing economic or technical conditions.
- For this reason, many platforms strive for a decentralized upgrade process, often mediated by a governance token.

2. DeFi Primitives - Fungible Tokens

Governance tokens

- A governance token can be implemented in many ways—with a static supply, an inflationary supply, or even a deflationary supply.
- A static supply is straightforward: purchased shares would correspond directly to a certain percentage control of the vote.
- MKR is an example of a static supply
- COMP is an example of inflationary supply to incentive use of the platform

2. DeFi Primitives - Fungible Tokens

Feature	Equity Tokens	Utility Tokens	Governance Tokens
Meaning	Represent ownership or stake in a company/project	Provide access to a product or service	Give voting power in a decentralized system
Primary Purpose	Investment & profit sharing	Functional use within platform	Decision-making & protocol control
Ownership Rights	Yes (similar to shares)	No ownership rights	No ownership, only governance rights
Voting Rights	May include shareholder voting	No	Yes (protocol proposals & changes)
Profit / Dividend	Yes (dividends, revenue share possible)	No profit rights	No direct profit (may affect ecosystem value)

2. DeFi Primitives - Fungible Tokens

Feature	Equity Tokens	Utility Tokens	Governance Tokens
Value Depends On	Company performance & profits	Demand for platform service	Influence over project direction
Legal Status	Usually treated as securities	Usually treated as non-securities	Regulatory status unclear (often utility-like)
Risk Level	Medium–High (market + business risk)	Medium (platform success risk)	High (protocol success & adoption)
Transferability	Transferable with restrictions	Freely transferable	Freely transferable
Example Use Case	Tokenized company shares	Tokens used to pay for cloud storage or gaming	DAO voting tokens

2. DeFi Primitives - Fungible Tokens

Feature	Equity Tokens	Utility Tokens	Governance Tokens
Typical Holders	Investors	Users & customers	Community members
Regulatory Control	High (subject to securities laws)	Low–Medium	Medium (depends on jurisdiction)
Economic Incentive	Capital appreciation & dividends	Platform usage benefit	Power & influence over protocol

2. DeFi Primitives - Non-Fungible Tokens

ERC-721

- ERC-721 defines the non-fungible standard.
- It is similar to ERC-20 except that each unit has its own unique ID
- Their alternate name, deeds, implies their use case as representing unique ownership of unitary assets; an example could be ownership of a particular P2P loan with its own rates and terms.
- Lottery tickets are nonfungible because only one or a limited number will be winning tickets and the remainder are worthless.
- NFTs can represent *collectibles* (e.g., ownership in a piece of art).

2. DeFi Primitives - Non-Fungible Tokens

Non-fungible tokens

ERC-721

- NFT's have reenergized the art market

NFT Sells for \$69.3 million,
most expensive ever



2. DeFi Primitives - Non-Fungible Tokens

ERC-1155

- ERC-20 and ERC-721 tokens require an individual contract and address deployed to the blockchain.
- These requirements can be cumbersome for systems that have many tokens, which are closely related, possibly even a mix of fungible and nonfungible token types.
- ERC-1155 resolves this complexity by defining a multi-token model in which the contract holds balances for a variable number of tokens, which can be fungible or nonfungible.

2. DeFi Primitives - Custody

Escrow

- A critical DeFi primitive is the ability to escrow or custody funds directly in a smart contract.
- This is different from the situation in ERC-20 when operators are approved to transfer a user's balance. In that case, the user still retains custody of his funds and could transfer the balance at any time or revoke the contract's approval.

2. DeFi Primitives - Custody

Escrow opens up new capabilities

- Additional primitives are possible:
 - Retaining fees and disbursing incentives
 - Facilitation of token swaps
 - Market making of a bonding curve
 - Collateralized Loans
 - Auctions
 - Insurance funds

2. DeFi Primitives - Custody

Escrow opens up new risks

- Users must exercise caution when sending tokens to contracts because the tokens could become permanently custodied if the contract has no encoded mechanism for releasing the funds of that particular token.

2. DeFi Primitives - Supply Adjustment

Burn (reduce supply)

- To burn a token means to remove it from circulation.
- Burning a token can take two forms:
 - Manually send a token to an unowned Ethereum address.
 - More efficient is to create a contract that is incapable of spending them.
- Either approach renders the burned tokens unusable, although the decrease in circulating supply would not be “known” by the token contract. Burning is analogous to the destruction or irreversible loss of currency in the traditional finance world, which is unknown to the issuing government.

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2. DeFi Primitives - Supply Adjustment

Burn mistakes

- In practice, ETH or ERC-20 tokens have frequently and accidentally been burned using both forms.
- Checksums are one method used to prevent accidental burn.
 - These are cryptographic primitives used to verify data integrity.
 - In the context of Ethereum addresses, [EIP-55](#) proposed a specific checksum encoding of addresses to stop incorrect addresses' receiving token transfers.
 - If an address used for a token transfer does not include the correct checksum metadata, the contract assumes the address was mistyped and the transaction would fail.

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2. DeFi Primitives - Supply Adjustment

Why burn?

- Here are some practical reasons:
 - Represent exiting of a pool and redemption of underlying (common in equity tokens like cTokens for Compound)
 - Increase scarcity to drive the price upward (e.g., AAVE)
 - Penalize bad acting

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2. DeFi Primitives - Supply Adjustment

Minting (increase supply)

- Minting increases the number of tokens in circulation.
- Contrary to burning, there is no mechanism for accidentally or manually minting tokens.
- Any mint mechanics have to be directly encoded into the smart contract mechanism.
- There are many use cases for minting as it can incentivize a wider range of user behavior.

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2. DeFi Primitives - Supply Adjustment

Minting (increase supply)

- Here are some examples:
 - Represent entering a pool and acquiring corresponding ownership share (common in equity tokens like cTokens for Compound)
 - Decrease scarcity (increase supply) to drive the price downward (seigniorage Stablecoin models like Basis/ESD)
 - Reward user behavior

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2. DeFi Primitives - Supply Adjustment

Minting as an incentive mechanism

- *Inflationary rewards* has become a common practice to encourage actions such as supplying liquidity or using a particular platform.
- Many users engage in yield farming, taking actions to seek the highest possible rewards. Platforms can bootstrap their networks by issuing a token with an additional value proposition in their network.
- Users can keep the token or sell it for a profit. Either way, utilization of the token benefits the platform by increasing activity.

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2. DeFi Primitives - Supply Adjustment

Bonding curves

- One advantage of being able to adjust supply up and down on a contractual basis is being able to define a bonding curve.
- A bonding curve is the price relationship between the token supply and a corresponding asset used to purchase the token(s).
- In most implementations investors sell back to the curve using the same price relationship.
- The relationship is defined as a mathematical function or as an algorithm with several clauses.

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2. DeFi Primitives - Supply Adjustment

Linear bonding curves

- Let TKN to denote the price of a token denominated in ETH (which could be any fungible cryptoasset) and use S to represent the supply.
- The simplest possible bonding curve would be $TKN=1$ (or any constant).
- This algorithmically enforces a one to one peg between ETH and TKN

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2. DeFi Primitives - Supply Adjustment

Linear bonding curves

- Next, consider a simple linear bonding curve, where m and b represent the slope and intercept, respectively, in a standard linear function.
- If $m = 1$ and $b = 0$, the first TKN would cost 1 ETH, the second would cost 2 ETH, and so on.
- A monotonically increasing bonding curve rewards early investors, because any incremental demand beyond their purchase price would allow them to sell back against the curve at a higher price point.

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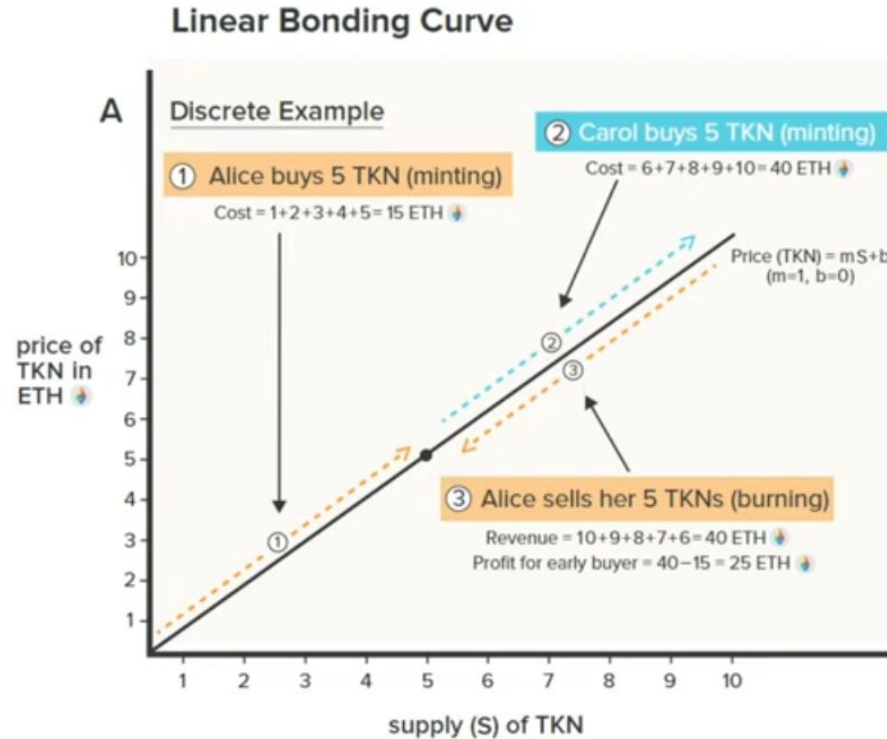
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2. DeFi Primitives - Supply Adjustment

Supply adjustment

Linear bonding curves

- Alice is rewarded for being an early investor



2. DeFi Primitives - Supply Adjustment

Linear bonding curves mechanics

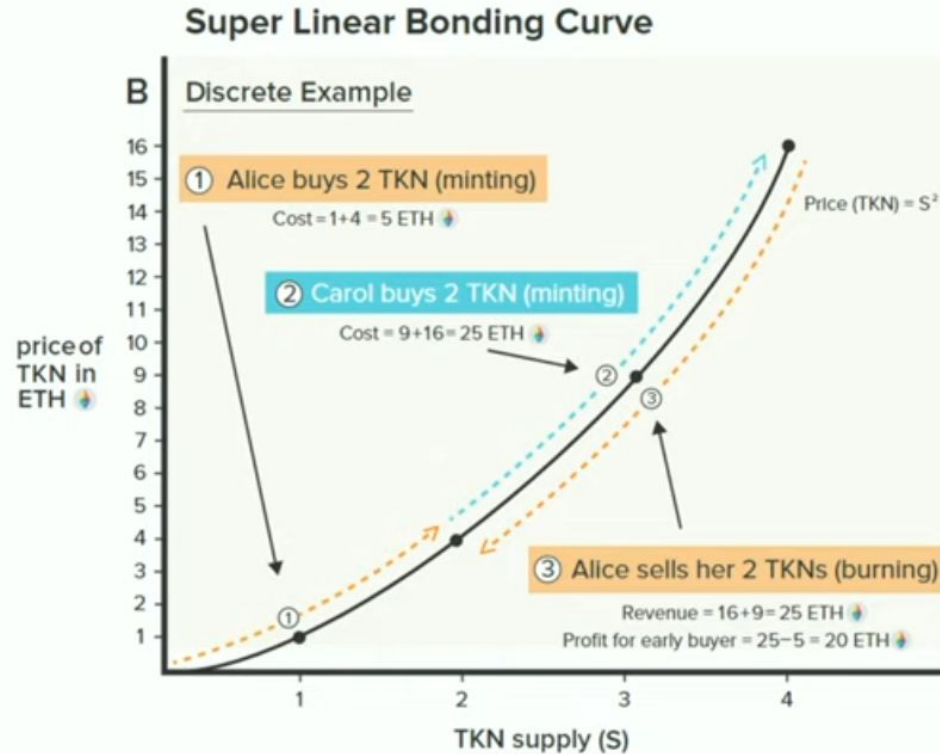
- The curve can be represented as a single smart contract with options for purchasing and selling the underlying token.
- The token to be sold can have either an uncapped supply with the bonding curve as an authorized minter or a predetermined maximum supply that is escrowed in the bonding curve contract.
- As traders purchase the token, the bonding curve escrows the incoming funding for the point in the future when a trader may want to sell back against the curve.

2. DeFi Primitives - Supply Adjustment

Supply adjustment

Super-linear bonding curves

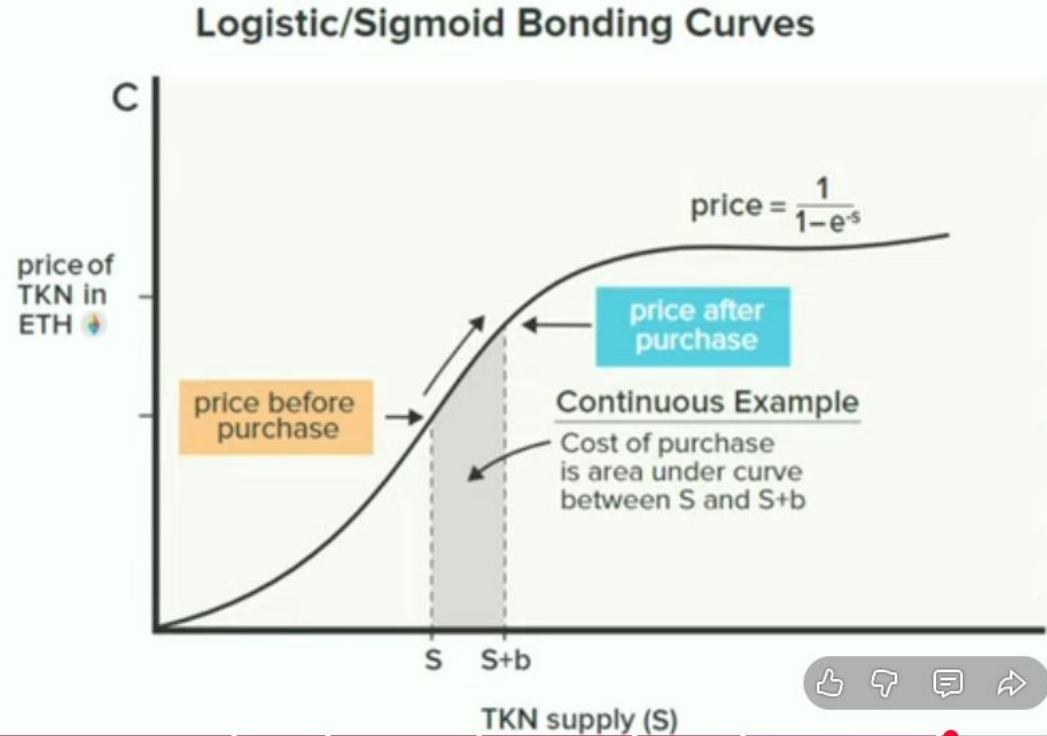
- Example: $TKN = S^2$
- More extreme rewards for early investors



2. DeFi Primitives - Supply Adjustment

Logistic bonding curves

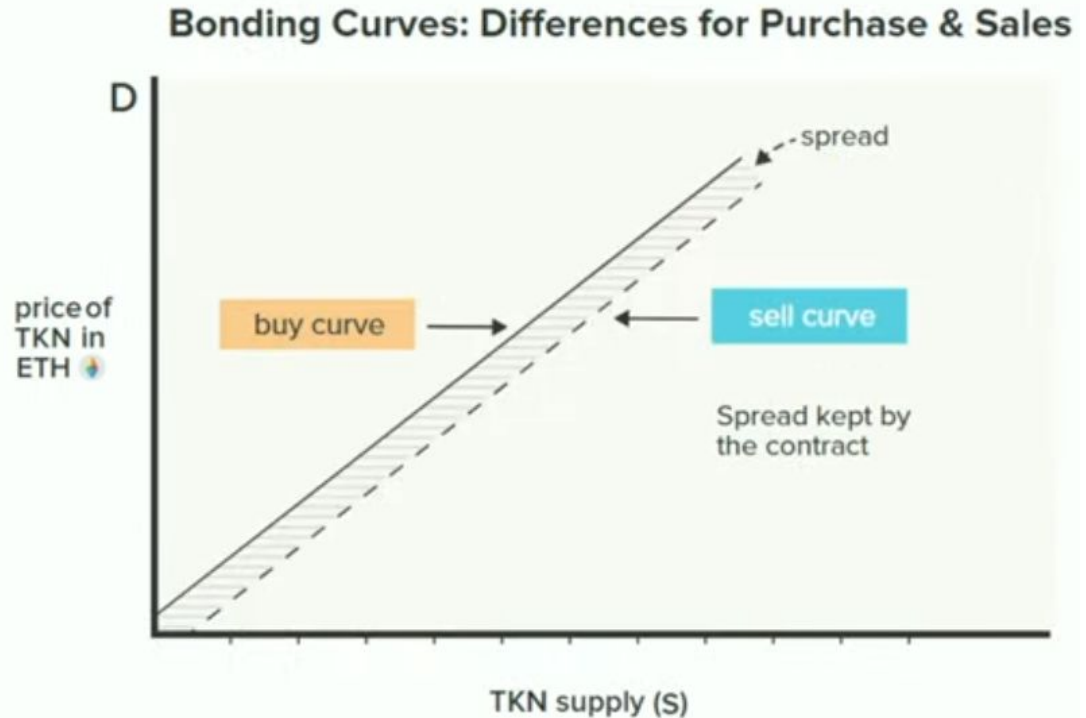
- Rewards early but then flattens out



2. DeFi Primitives - Supply Adjustment

Buy vs. sell bonding

- It is possible to have different curves for buying and selling
- The spread is kept by the contract



2. DeFi Primitives - Supply Adjustment

Feature	Linear Bonding Curve	Super-Linear Bonding Curve	Logistic Bonding Curve
Definition	Token price increases at a constant rate as supply increases	Token price increases faster than linear as supply increases	Token price follows an S-shaped curve (slow → fast → slow growth)
Price–Supply Relationship	Price $\propto$ Supply	Price $\propto$ Supply ² (or higher power)	Price grows slowly at first, rapidly in mid-range, then slows near max supply
Initial Price Growth	Moderate and predictable	Slow at first, then steep	Very slow at start
Middle Phase	Steady growth	Rapid price escalation	Very fast growth
Later Phase	Continues steadily	Extremely expensive	Price stabilizes (approaches upper limit)
Supply Control	Weak control over oversupply	Strong control (discourages large minting)	Strong natural cap due to saturation

2. DeFi Primitives - Supply Adjustment

Feature	Linear Bonding Curve	Super-Linear Bonding Curve	Logistic Bonding Curve
Volatility	Low–Medium	High	Medium
User Incentive	Fair for early and late buyers	Strong incentive for early buyers	Balanced incentive (early & mid adopters)
Risk of Speculation	Medium	Very high	Lower than super-linear
Complexity	Simple	Medium	Complex
Typical Formula	$P = aS + b$	$P = aS^2 + b$	$P = L / (1 + e^{-k(S-S_0)})$
Best Use Case	Stable token distribution	Scarcity-driven assets	Adoption-based ecosystems
Example Use	Basic DAO tokens	Investment-style tokens	Social tokens, community tokens

2. DeFi Primitives - Incentives

Incentives

Types of incentives

- Two categories of incentives: staked incentives and direct incentives
 - Staked incentives apply to a balance of tokens custodied in a smart contract.
 - Direct incentives apply to users within the system who do not have a custodied balance.

2. DeFi Primitives - Incentives

Incentives

Staking rewards

- A *staking reward* is a positive staked incentive by which a user receives a bonus in his token balance based on the stake she has in the system.
- Several verticals of incentive customization are possible:
 - Stake requirement options:
 - minimum threshold or applied to all staked balances on a pro rata basis
 - Reward options:
 - Fixed payout or pro rata payout
 - Same token type as staked or a distinct token

2. DeFi Primitives - Incentives

Incentives

Staking rewards examples

- The Compound protocol issues staking rewards on user balances that are custodied in a borrowing or lending position. These rewards are paid in a separate token (COMP) funded by custodied COMP, which has a fixed supply, and applied to all staked balances on a pro rata basis.
- The Synthetix protocol issues staking rewards on staked SNX, its protocol token which has unlimited supply. The rewards are paid in SNX, funded by inflation, and issued only if the user meets a minimum-collateralization-ratio threshold.

2. DeFi Primitives - Incentives

Incentives

Slashing

- *Slashing* is the removal of a portion of a user's staked balance, thereby creating a negative staked incentive.
- Slashing occurs as the result of an undesirable event.
- A *slashing condition* is a mechanism that triggers a slashing.

2. DeFi Primitives - Incentives

Incentives

Slashing

- Slashing customization
 - Removed funds options:
 - Complete or partial slashing
 - Slashing condition options:
 - Undercollateralization triggers liquidation
 - Detectable malicious behavior by user
 - Change in market conditions triggers necessary contraction

2. DeFi Primitives - Incentives

Slashing example

- With collateralized loans, one slashing mechanism is liquidation
- In a liquidation, potential liquidators receive a direct incentive to execute the liquidation through auctioning or directly selling the collateral; the balance of funds remaining after the liquidation stays with the original owner.

2. DeFi Primitives - Incentives

Direct rewards and keepers

- *Direct rewards* are positive incentives that include payments or fees associated with user actions.
- Ethereum interactions begin with a transaction, and all transactions begin with an externally owned account.
- An EOA, whether controlled by a human user or an off-chain bot, is (importantly) off chain.
- Thus autonomous monitoring of market conditions is either expensive (costs gas) or technically infeasible.
- As a result, no transaction happens automatically on Ethereum without being purposely set in motion.

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2. DeFi Primitives - Incentives

Direct rewards example

- The classic example of a transaction that must be set in motion is when a collateralized debt position becomes undercollateralized.
- This use case does not automatically trigger a liquidation; the EOA must trigger the liquidation.
- For this use case and others, EOAs generally receive a direct incentive to trigger the contract.
- The contract then evaluates the conditions and liquidates or updates if everything is as expected.

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2. DeFi Primitives - Incentives

Incentives

Keeper

- A *keeper* is a class of EOA incentivized to perform an action in a DeFi protocol or other dApp.
- A keeper is rewarded by receiving a fee, either flat or percentage of the incented action.
- Keeper rewards may also be structured as an auction to ensure competition and best price.
- Keeper auctions are very competitive because the information available in the system is almost entirely public.

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2. DeFi Primitives - Incentives

Incentives

Keeper downside

- A side effect of direct rewards for keepers is that gas prices can inflate due to the competition for these rewards.
- That is, more keeper activity generates additional demand for transactions, which in turn increases the price of gas.

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2. DeFi Primitives - Incentives

Incentives

Fees

- Fees are typically a funding mechanism for the features of the system or platform.
- They can be flat or percentage based, depending on the desired incentive. Fees can be imposed as a direct negative incentive or can be accrued on staked balances.
- Accrued fees must have an associated staked balance to ensure the user pays them.

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2. DeFi Primitives - Incentives

Incentives

Fees

- Given the pseudonymous anonymous nature of Ethereum accounts—all that is known about an Ethereum user is his wallet balance and interactions with various contracts on Ethereum—the imposition of fees is a design challenge.
- If the smart contract is open to any Ethereum account, the only way to guarantee off-chain enforcement or legal intervention is for all debts to be backed by staked collateral, which is transparent and enforceable.
- The challenges created by anonymity make other mechanisms, such as reputation, unsuitable alternatives to staked balances.

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2. DeFi Primitives - Swaps and Loans

- A swap is simply the exchange of one type of token to another.
- There are a number of ways to do this
- Most use a centralized exchange like Coinbase or Coinbase Pro

All assets ▾
 1D ▾
 INR ▾
 10 rows ▾

	Asset	Market price	Chart	Change	Mkt cap	Volume
☆	 Bitcoin BTC	₹6,274,366.81		↘ 0.70%	₹125.8T	₹4.5T
☆	 Ethereum ETH	₹181,872.74		↘ 2.10%	₹22.0T	₹2.2T
☆	 Tether USDT	₹92.03		↗ 0.01%	₹16.9T	₹8.7T
☆	 BNB BNB	₹58,274.19		↘ 0.72%	₹8.0T	₹164.3B

www.coinbase.com/en-in/price/lighter

Tradable	Top gainers	New on Coinbase
	Bitcoin	₹6,253,076.70 ↘ 1.22%
	Ethereum	₹181,131.33 ↘ 2.76%
	Tether	₹92.02 ↗ 0.02%
	BNB	₹58,061.48 ↘ 1.38%
	XRP	₹124.66 ↘ 2.46%
	USDC	₹92.01 ...

2. DeFi Primitives - Swaps and Loans

Orderly Cryptocurrency?

- **“Arbitrageable”**: Cryptocurrencies have low transaction costs, globally fungible
 - In theory, they should have tight spreads, low volume, orderly markets
- **Instead:**
 - BTC prices differ by hundreds of dollars across exchanges
 - BTC daily volume of \$25bn, 15% of entire market cap
 - Annual volume is 55x the market cap
 - Annual volume of Apple is only 1.7x market cap
 - How can this be?

#	Source	Pair	Volume (24h)	Price
1	 Coinsbit	BTC/USDT	\$878,944,892	\$9,394.74
2	 Bilaxy	BTC/USDT	\$813,087,459	\$9,401.63
3	 BHEX	BTC/USDT	\$739,387,382	\$9,397.50
4	 CoinBene	BTC/USDT	\$724,065,563	\$9,398.71
5	 Folgory	BTC/USDT	\$637,148,703	\$9,390.86
6	 EXX	BTC/USDT	\$601,369,387	\$9,398.71
7	 BitForex	BTC/USDT	\$573,580,912	\$9,398.63
8	 Hotbit	BTC/USDT	\$526,715,554	\$9,397.91
9	 Coinel	BTC/USDT	\$513,734,235	\$9,398.71
10	 ChainX	BTC/KRW	\$506,932,884	\$9,042.87

Source: CoinMarketCap.com

2. DeFi Primitives - Swaps and Loans

Fake Exchanges

Why would exchanges exaggerate volume

- **Initial Coin and Exchange Offerings (ICOs/IEOs):** Showing up at the top of these lists can attract coins to list on your exchange
- **"Fake it till you make it":** Crypto traders may be attracted to trade on your exchange, thereby increasing the "true" volume
- **Over 90% of trading volume is fake!**
 - Fake transactions can either occur off-chain (within addresses internally) or on-chain (pay transaction and show up on blockchain)
 - How can we identify a fake exchange?

2. DeFi Primitives - Swaps and Loans

What a Normal Exchange Looks Like



2. DeFi Primitives - Swaps and Loans

What a Normal Exchange Looks Like

Key Elements

- **Order Book:** List of trades (size and prices) that have been submitted but not yet filled
 - Orders in red are to sell; in green are to buy
 - Spread is given by difference between minimum sell price and maximum buy price
 - Trades occur when someone is willing to sell at a price at which someone wants to buy
 - Example: 5.601 BTC being offered for sale at \$10,336.30 per BTC. Buyer wants to purchase 0.4045 BTC at \$10,335.05. The spread between the cheapest sale price and highest purchase price is \$1.19.
- **Low spread:** The spread of BTC is \$1.19 (only 0.01% of the price)
 - Spread of Apple is \$0.15 (0.05% of price)
- **Round numbers:** Many orders are for round numbers of BTC, evidence of a normal exchange

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2. DeFi Primitives - Swaps and Loans

What a Normal Exchange Looks Like

Key ingredients

- **Price History:** Red and green bars reflecting price movements every interval (typically 5-10 minutes)
 - Notice that it is again streaky
- **Trading Volume:** Gray bars at the bottom show amount of BTC traded each interval
 - Notice randomness in trading volume: some intervals have higher volume than others (i.e., slump in volume after 5:00pm)



Source: Coinbase.com

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[illegible]

2. DeFi Primitives - Swaps and Loans

What a Fake Exchange Looks Like

Examples

- **RightBTC**: an exchange with 4x the volume of Coinbase Pro
- **Hours of zero trading volume**: There are large gaps (12- or even 24-hour periods) without any trading volume.



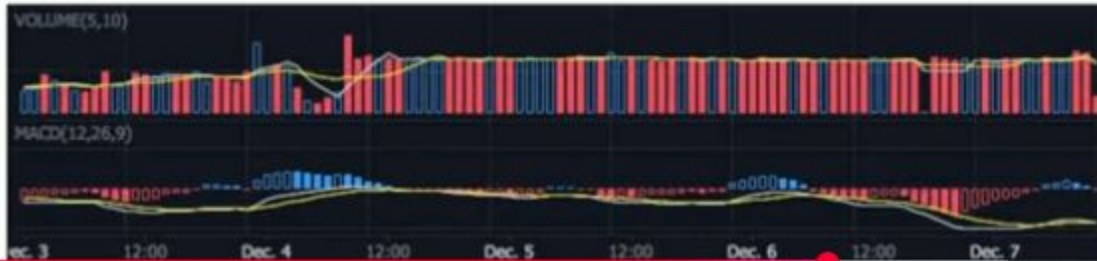
Source: Bitwise Asset Management

2. DeFi Primitives - Swaps and Loans

What a Fake Exchange Looks Like

Examples

- **CHAOEX**: an exchange with roughly the same volume as Coinbase Pro
- **Constant volume**: Note that the volume is basically constant for over 3 days in this case
 - Volume is insensitive to market hours, news, weekends, etc.

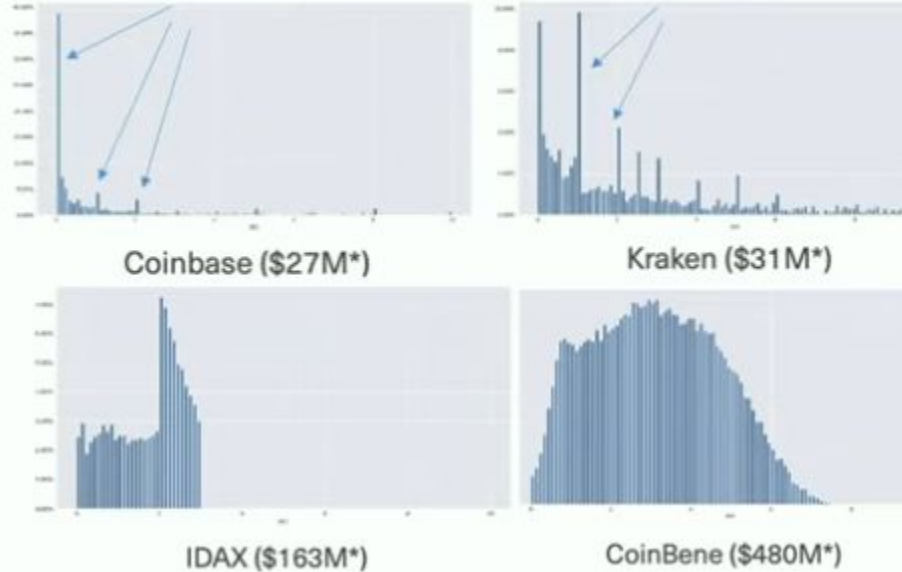


2. DeFi Primitives - Swaps and Loans

Trade Size of Real vs. Fake Exchanges

Huge Differences!

- Real exchanges have very few large trades, cluster around integer values
- Fake exchanges have very strange distributions
 - Random spikes, very dispersed volume, etc.



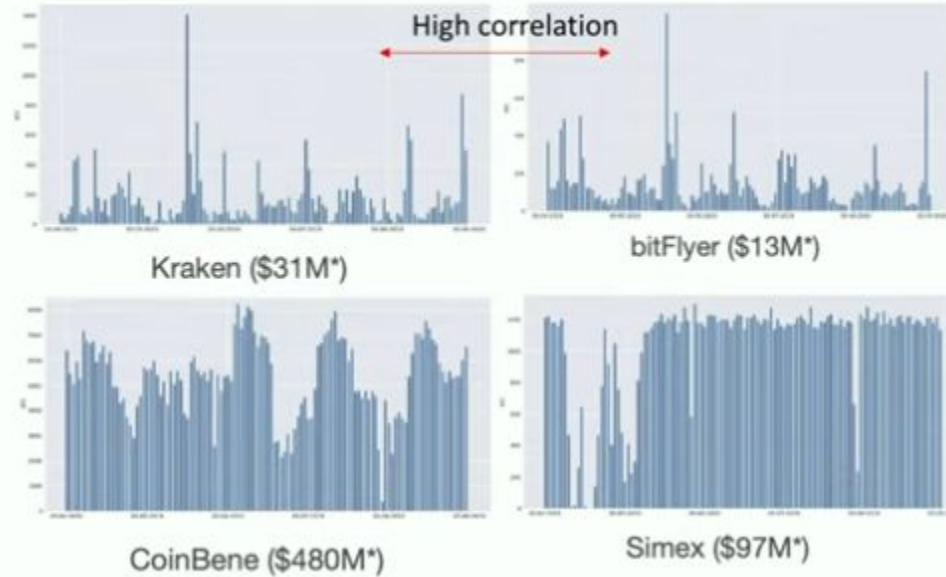
Source: Bitwise Asset Management

2. DeFi Primitives - Swaps and Loans

Trade Volume of Real vs. Fake Exchanges

Huge Differences!

- Real exchanges have similar spikes in volume at the same time, rise and fall daily
- Fake exchanges have volume that is extremely high almost constantly



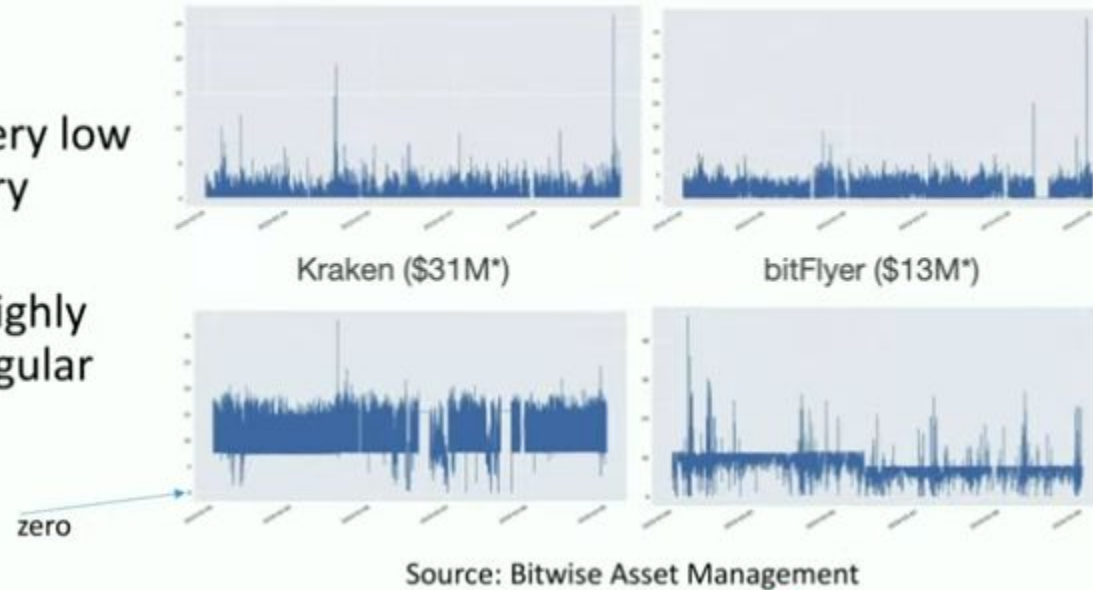
Source: Bitwise Asset Management

2. DeFi Primitives - Swaps and Loans

Spreads of Real vs. Fake Exchanges

Huge Differences!

- Real exchanges have very low spreads that do not vary much and rarely peak
- Fake exchanges have highly volatile spreads and regular peaks



2. DeFi Primitives - Swaps and Loans

How Much Volume is Fake

- **Only 4.5% of Total Volume is Real:** This reduces BTC daily volume from \$25bn “fake” volume to just over \$1bn actual volume
 - Annual “real” volume is 1.7x market cap
 - Same as Apple’s turnover
- Cryptocurrency markets look much more orderly when correcting for fake volume
- Being able to distinguish fake volume crucial to understanding centralized exchange

2. DeFi Primitives - Swaps and Loans

Swap

What is a swap?

- A swap is simply the exchange of one type of token to another.
- The key benefit of swapping in DeFi is that it is atomic and noncustodial.
- Funds can be custodied in a smart contract with withdrawal rights that can be exercised at any time before the swap is completed.
- If the swap does not complete, all parties involved retain their custodied funds.

2. DeFi Primitives - Swaps and Loans

Swap

What is a swap?

- The swap only executes when the exchange conditions are agreed to and met by all parties, and are enforced by the smart contract.
- If any condition is not met, the entire transaction is cancelled. A platform that facilitates token swapping on Ethereum in a noncustodial fashion is a *decentralized exchange* (DEX).
- There are two primary mechanisms for DEX liquidity: one is an order-matching approach and the other is an *Automated Market Maker*.

2. DeFi Primitives - Swaps and Loans

Swap

Order book matching

- *Order-book matching* is a system in which all parties must agree on the swap exchange rate.
- Market makers can post bids and asks to a DEX and allow takers to fill the quotes at the pre-agreed-upon price.
- Until the offer is taken, the market maker retains the right to remove the offer or update the exchange rate as market conditions change.

2. DeFi Primitives - Swaps and Loans

Swap

Order book matching

- A leading example of a fully on-chain order book is [Kyber](#).
- “KyberSwap is a non custodial platform. It means you are in total control of your funds. In a typical centralized exchange - Before placing any trade, you are first required to deposit your funds to exchange. In KyberSwap you do not need to deposit any funds. Just connect your Ethereum wallet and place a trade directly from your wallet.”



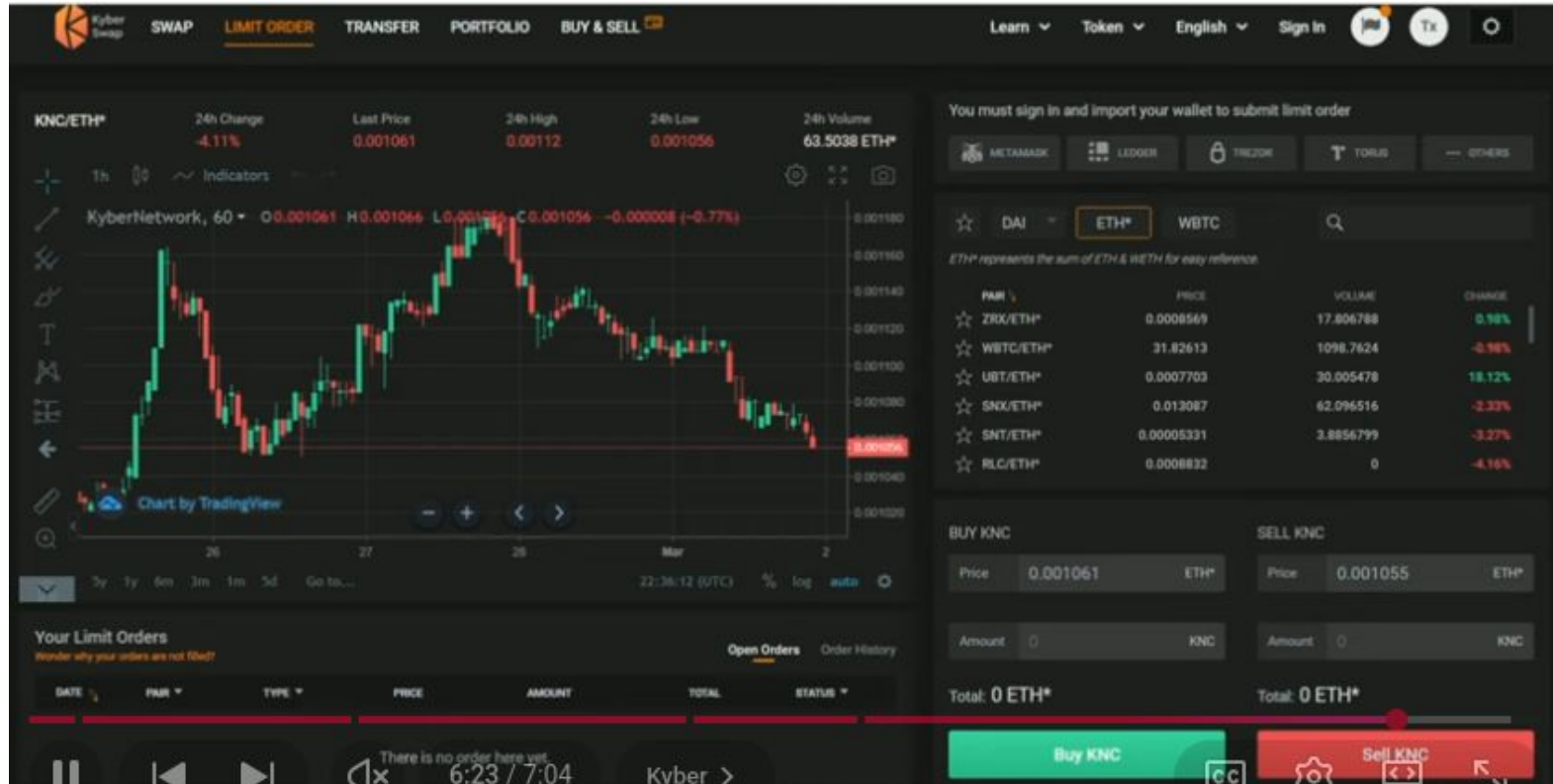
Seamless Token Swaps, Anywhere

Kyber is a blockchain-based liquidity protocol that aggregates liquidity from a wide range of reserves, powering instant and secure token exchange in any decentralized application.

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BT

2. DeFi Primitives - Swaps and Loans



2. DeFi Primitives - Swaps and Loans

Swap

Order book matching issues

- The order-matching approach is expensive and inefficient because each update requires an on-chain transaction.
- An insurmountable inefficiency with an order-book matching is that both counterparties must be willing and able to exchange at the agreed-upon rate for the trade to execute.
- This requirement creates limitations for many smart contract applications in which demand for exchange liquidity cannot be dependent on a counterparty's availability.

2. DeFi Primitives - Swaps and Loans

Swap

Automated Market Makers (AMMs)

- An Automated Market Maker (AMM) is a smart contract that holds assets on both sides of a trading pair and continuously quotes a price for buying and for selling.
- Based on executed purchases and sales, the contract updates the asset size behind the bid and the ask and uses this ratio to define its pricing function.
- The contract can also take into account more complex data than relative bid/ask size when determining price.
- From the contract's perspective, the price should be risk-neutral where it is indifferent to buying or selling.

2. DeFi Primitives - Swaps and Loans

Swap

Naïve AMM

- A naive AMM might set a fixed price ratio between two assets.
- With a fixed price ratio, when the market price shifts between the assets, the more valuable asset would be drained from the AMM and arbitrated on another exchange where trading is occurring at the market price.
- The AMM should have a pricing function that can converge on the market price of an asset so that it becomes more expensive to purchase an asset from the trading pair as the ratio of that asset to the others in the contract decreases.

2. DeFi Primitives - Swaps and Loans

Swap

Advantages of AMM

- Main benefit is the constant availability 24/7 and that a traditional counterparty is not necessary to execute a trade.
- These provisions are very important for smart contracts and DeFi development because of the guarantee that a user can exchange assets at any moment if necessary.
- A user maintains custody of her funds until she completes the trade, hence, counterparty risk is zero.

2. DeFi Primitives - Swaps and Loans

Swap

Composable liquidity of AMM

- An additional benefit is *composable liquidity*, which means any exchange contract can plug into the liquidity and exchange rates of any other exchange contract.
- AMMs make this particularly easy because of their guaranteed availability and their allowing one-sided trading against the contract.
- Composable liquidity fits with concept of DeFi Legos.

2. DeFi Primitives - Swaps and Loans

Swap

Impermanent loss of AMM

- One drawback to an AMM is the concept of *impermanent loss*, the opportunity-cost dynamic between offering assets for exchange and holding the underlying assets to potentially profit from the price movement.
- The loss is impermanent because it can be recovered if the price reverts to its original level.

Swap

Impermanent loss example

- Initial conditions in market:
 - Token A = 1 ETH and
 - Token B = 1 ETH
- AMM has an exchange rate of 1:1
- Contract has 100 A and 100 B.
So the total value of escrow is 200 ETH

AUTOMATED MARKET MAKER



Initial Conditions

Asset A = 1 ETH

Asset B = 1 ETH

Exchange rate in AMM = 1:1

AMM has 100 A and 100 B

Total escrow = 200 ETH

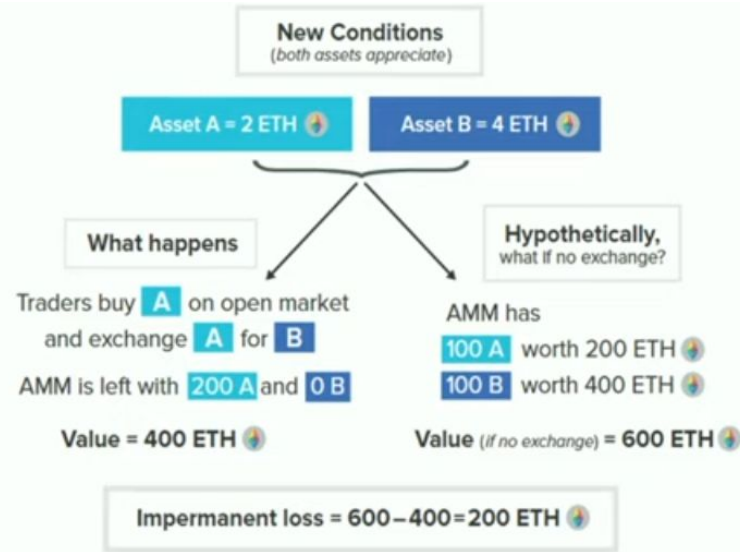


2. DeFi Primitives - Swaps and Loans

Swap

Impermanent loss example

- New conditions. Both tokens appreciate in value. Now:
 - Token A = 2 ETH and
 - Token B = 4 ETH
- AMM has an exchange rate of 1:1
- Traders buy token A on open market (like Coinbase) and exchange it in the AMM for B – draining all the B.



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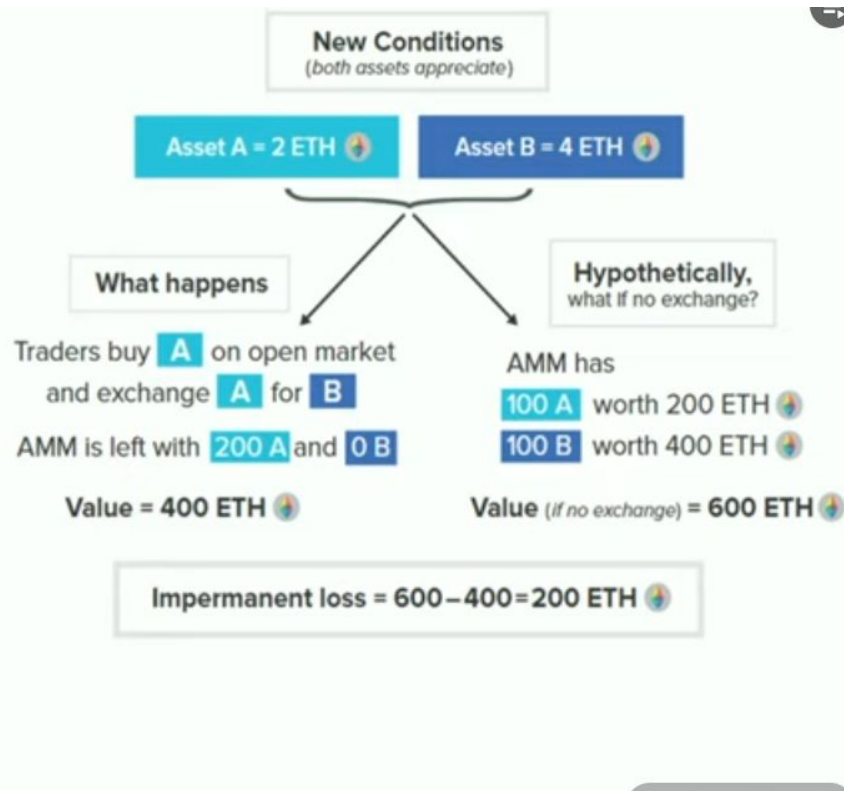
2. DeFi Primitives - Swaps and Loans

Swaps - Automated Market Makers

Swap

Impermanent loss example

- Contract left with 200 A and zero B.
- Value = 400 ETH
- However, if there was no exchange in the AMM, the value would be 600 ETH
- Impermanent loss is the difference $600 - 400 = 200$ ETH



2. DeFi Primitives - Swaps and Loans

Swap

Impermanent loss Uniswap v2

- Initial market prices are 1 ETH = 100 DAI
- Alicia deposits 1 ETH and 100 DAI into a liquidity pool
- Alicia will earn a fee for providing liquidity
- Notice that an equal value of DAI and ETH are deposited
- There are others like Alicia in the pool which has a total of 10 ETH and 1,000 DAI. The total liquidity is 10,000 = 10 x 1,000
- Alicia owns 10% of the pool

Swap

Impermanent loss Uniswap v2

- New market prices are $1 \text{ ETH} = 400 \text{ DAI}$
- Arbitrageurs see the opportunity and buy DAI in open market and use DAI to withdraw ETH. The exchange price depends on the ratio of price whereas the liquidity (10,000) remains constant.
- Arbitrageurs will drain 5 ETH so the pool now has 5 ETH and 2,000 DAI. Notice liquidity is still 10,000 and the new ratio is 1:400 (reflecting market prices).

2. DeFi Primitives - Swaps and Loans

Swap

Impermanent loss Uniswap v2

Vault	$x*y=k$	k is constant	
ETH	DAI	product	price of ETH in DAI
10	1,000.00	10,000	100.00
9	1,111.11	10,000	123.46
8	1,250.00	10,000	156.25
7	1,428.57	10,000	204.08
6	1,666.67	10,000	277.78
5	2,000.00	10,000	400.00

2. DeFi Primitives - Swaps and Loans

Swap

Impermanent loss Uniswap v2

- Alicia owns 10% and withdraws all her funds from the pool. That will be 0.5 ETH and 200 DAI.
- USD value is \$400 ($\$400 \times 0.5 + \1×200)
- Her original investment was \$200
- However, if she did not deposit into the pool, the value of the assets would have been \$500 ($\$400 \times 1 + \1×100)
- The impermanent loss is \$100 ($\$500 - \400)
- Note that there is a profit overall plus we are not accounting for the fees that Alicia would earn for providing liquidity

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2. DeFi Primitives - Swaps and Loans

Impermanent loss features

- Impermanent loss occurs for any shift in price and liquidity, because the contract is structured to sell the appreciating asset and to buy the depreciating asset.
- An important feature of impermanent loss is *path independence*. In our example, it is irrelevant whether 1 or 100 traders consumed all the liquidity.
- The final exchange rate and contract asset ratios yield the same impermanent loss regardless of the number of trades or the direction of the trades.
- Because of path independence, impermanent loss is minimized on trading pairs that have correlated prices (*mean-reverting pairs*).
- Thus, stablecoin trading pairs are particularly attractive for AMMs.

2. DeFi Primitives - Swaps and Loans

Collateralized loans

Role of debt and lending in DeFi

- Debt and lending are perhaps the most important financial mechanisms that exist in DeFi, and in traditional finance.
- Any loan of non-zero duration (e.g., foreshadowing flash loan) must be backed by an equivalent or excess amount of collateral.
- Requiring collateral contractually prevents a counterparty from defaulting.
- An uncollateralized mechanism raises the risk that a counterparty could steal funds, especially in an open and anonymous system such as Ethereum.

2. DeFi Primitives - Swaps and Loans

Collateralized loans

Foreclosure risk

- A risk of overcollateralized positions is that the collateral becomes less valuable than the debt, leading to a foreclosure without an option for recovery.
- Therefore, more-volatile types of collateral require larger collateralization ratios in order to mitigate this risk.

Collateralized loans

Liquidation

- To avoid liquidation it is imperative that debt remain overcollateralized by a margin sufficiently large that moderate price volatility does not place the collateral value in jeopardy.
- Smart contracts commonly define a minimum collateralization threshold below which the collateral can be liquidated and the position closed.
- The collateral could be auctioned or directly sold on a DEX, likely with an AMM, at the market price.

2. DeFi Primitives - Swaps and Loans

Collateralized loans

Liquidation trigger

- Positions in the Ethereum blockchain cannot be liquidated automatically, so an incentive is needed.
- The incentive often takes the form of a percentage fee allocated to an external **keeper** who is able to liquidate the position and collect the reward.
- Any remaining collateral is left to the original holder of the position.
- In some cases, the system will leave all remaining collateral to the keeper as a stronger incentive.
- Because the penalty for liquidation is high and most collateral types are volatile, platforms generally allow users to top up their collateral to maintain healthy collateralization ratios.

2. DeFi Primitives - Swaps and Loans

Collateralized loans

Collateralization can back a token

- An implication of collateralized loans and token supply adjustment is that collateralization can back the value of a synthetic token.
- The synthetic token is an asset created and funded by a debt, which is the requirement to repay the synthetic token in order to reclaim the collateral.
- The synthetic token can have a utility mechanism or represent a complex financial derivative, such as an option or bond (e.g., Synthetix Synth and Yield yToken). A stablecoin that tracks the price of an underlying asset can also be a synthetic token of this type (e.g., MakerDAO DAI).

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Flash loans

Traditional finance

- A financial primitive that uniquely exists in DeFi and dramatically broadens certain types of financial access is a *flash loan*.
- In traditional finance, a lender is compensated for providing the capital and bearing the risk of default by the interest amount charged over the life of the loan.
- The interest rate is typically higher the longer the duration of the loan, because the longer time to repay exposes the lender to greater risk that the borrower may default.

2. DeFi Primitives - Swaps and Loans

Flash loans

Zero-duration loans

- Reversing the concept leads to the conclusion that shorter-term loans should be less risky and therefore require less compensation for the lender.
- A flash loan is an instantaneous loan paid back within the same transaction.
- A flash loan is similar to an overnight loan in traditional finance, but with a crucial difference—repayment is required within the transaction and enforced by the smart contract.

2. DeFi Primitives - Swaps and Loans

Flash loans

Risk of flash loans

- A thorough understanding of an Ethereum transaction is important for understanding how flash loans work.
- One clause in the transaction is vital: if the loan is not repaid with required interest by the end of the transaction, the whole process reverts to the state before any money ever left the lender's account.
- In other words, either the user successfully employs the loan for the desired use case and completely repays it in the transaction or the transaction fails and everything resets as if the user had not borrowed any money.

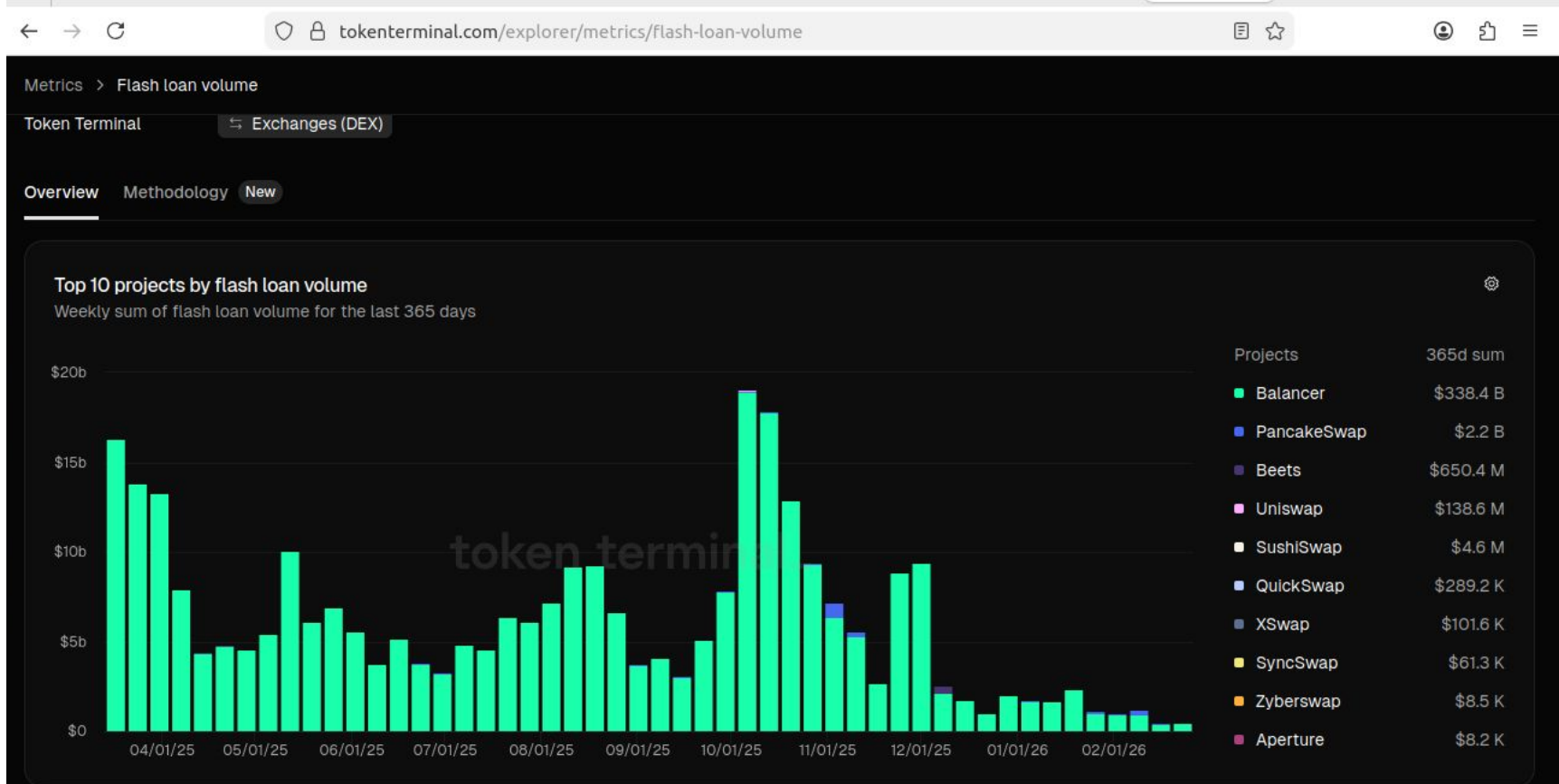
2. DeFi Primitives - Swaps and Loans

Flash loans

Risk of flash loans

- Flash loans essentially have zero counterparty risk or duration risk.
- They allow a user to take advantage of arbitrage opportunities or refinance loans without pledging collateral.
- This capability allows anyone in the world to have access to opportunities that typically require very large amounts of capital investment.
- This type of innovations that cannot exist in the world of traditional finance.
- However, these are not “risk free” because of smart contract risk.

2. DeFi Primitives - Swaps and Loans



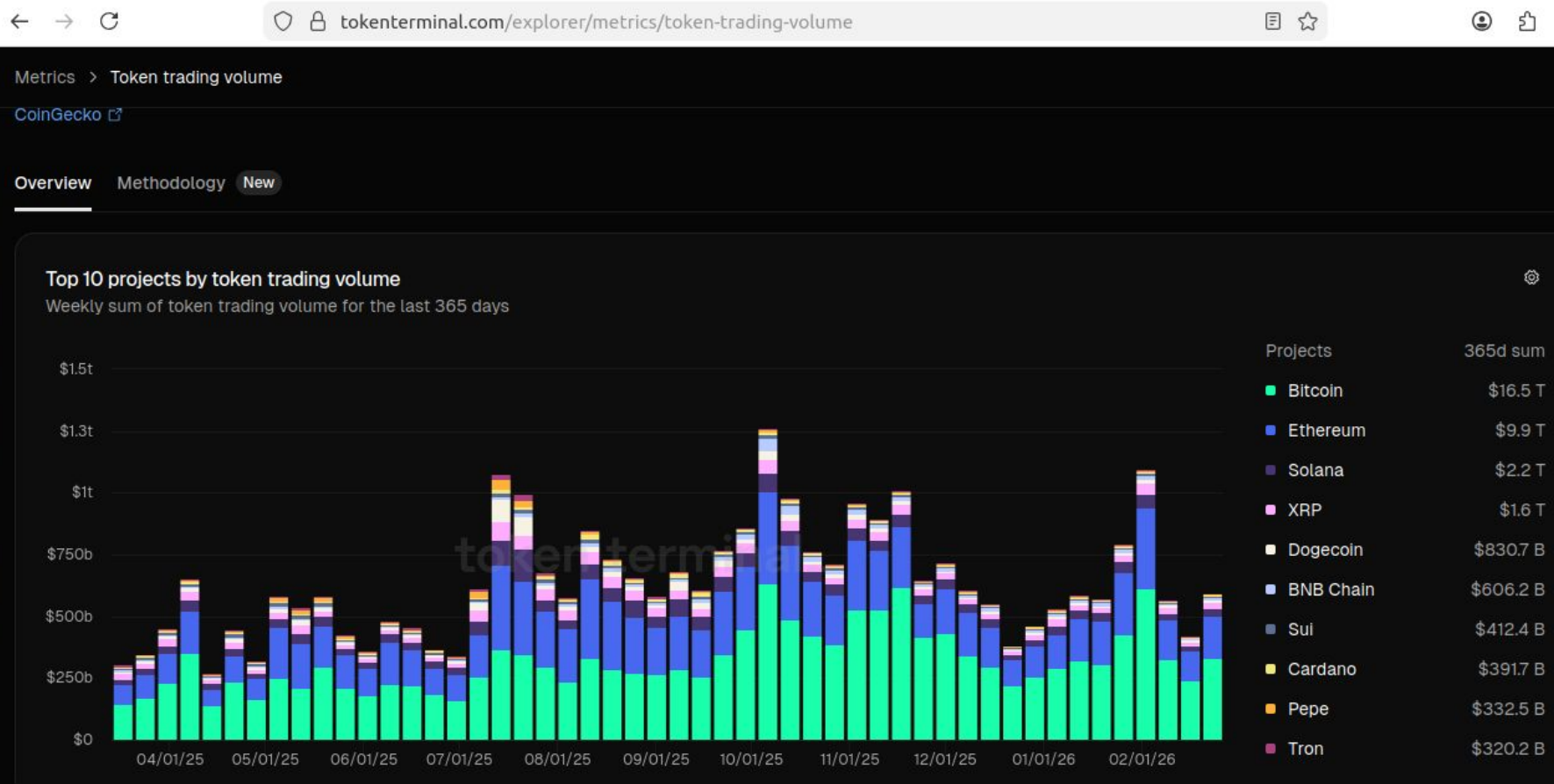
2. DeFi Primitives - Swaps and Loans

Feature	Centralized Exchange (CEX)	Decentralized Exchange (DEX)
Control of Funds	Exchange holds your funds	You control your wallet
Custody	Custodial	Non-custodial
KYC Required	Usually Yes	Usually No
Speed	Fast (off-chain matching)	Slower (on-chain transactions)
Fees	Trading fees	Gas fees + protocol fee
Security Risk	Risk of hacks / company failure	Smart contract risk
Privacy	Lower (identity verification)	Higher
Ease of Use	Beginner-friendly	Requires wallet knowledge
Token Listings	Limited (approved tokens)	Open listing (any token)
Customer Support	Yes	No central support

3. DeFi Key Metrics:

- measure ecosystem growth and health
- Investors and analysts use metrics to evaluate projects
- Key metrics:
 - Total Value Locked (TVL) → Measures liquidity strength
 - Daily Active Users (DAU) → Measures user adoption
 - Market Cap → Measures valuation
- These indicators [help assess adoption, liquidity, and valuation](#)

3. DeFi Key Metrics:



3. DeFi Key Metrics: Total Value Locked (TVL)

- [Total capital locked in DeFi protocols](#)
- Includes lending, staking, liquidity pools
- Indicates liquidity and protocol trust
- Higher TVL = stronger ecosystem confidence

Benefits

- Measures capital efficiency
- Attracts developers and investors
- Used to compare competing protocols
- Important for yield farming and staking analysis

Limitation: [Can be inflated by token price volatility](#)

3. DeFi Key Metrics: Daily Active Users (DAU)

- [Number of unique wallet addresses interacting daily](#)

Significance

- Reflects real adoption and engagement
- Helps measure platform growth
- Indicates community strength

Limitation: [One user may control multiple wallets](#)

3. DeFi Key Metrics: Market Capitalization

- Formula: [Token Price × Circulating Supply](#)
- Represents total network valuation
- Used to rank DeFi projects
- Large-cap = more stability
- Small-cap = higher growth potential, higher risk

4. DeFi Major Business Models:

- Decentralized Currencies → Medium of exchange & value storage
- Decentralized Payment Services → Transaction processing services
- Decentralized fundraising → Capital formation mechanisms
- Decentralized Contracting → Automated financial agreements

4. DeFi Major Business Models: Decentralized Currencies

- operates without central authority
- Secured by blockchain networks
- Used for payments, store of value, trading
- Examples: Native blockchain tokens and stablecoins
- Revenue model: transaction fees, ecosystem growth



4. DeFi Major Business Models: Decentralized Currencies

Key Characteristics

- Built on blockchain technology
- Peer-to-peer transactions
- Transparent and immutable ledger
- Cryptographically secured

Examples

- Bitcoin
- Ethereum
- Stablecoins like USDC

Business Model Logic

- Value derived from network adoption
- Revenue via transaction fees
- Token appreciation increases ecosystem value

Advantages

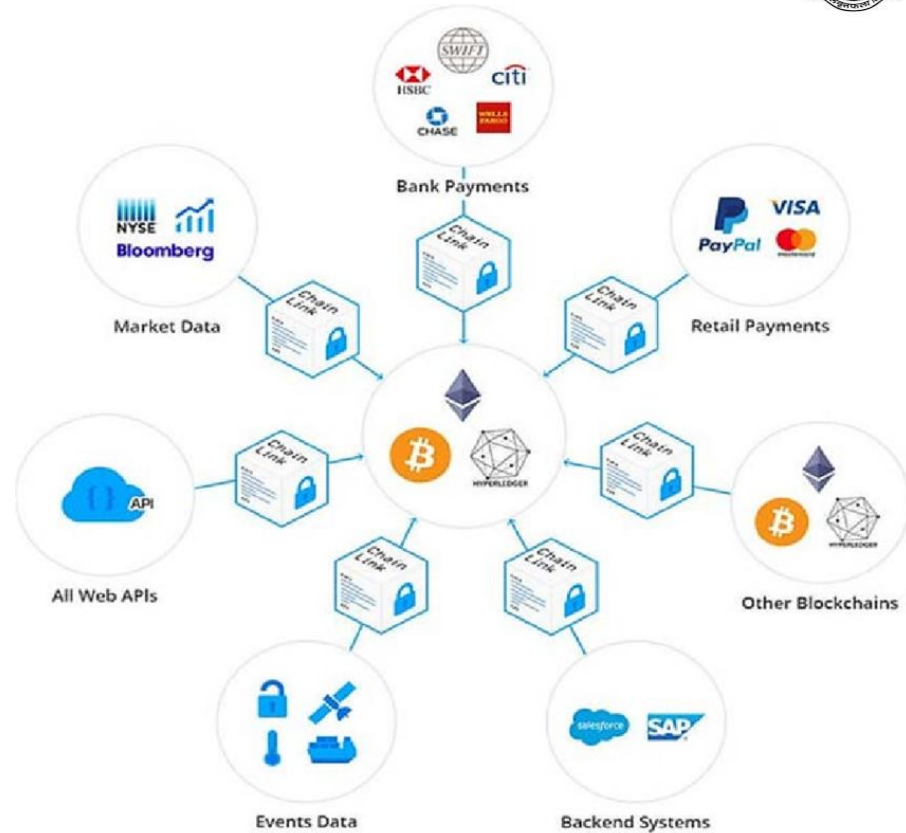
- ✓ No central control
- ✓ Global accessibility
- ✓ Lower transaction barriers

Challenges

- Price volatility
- Regulatory uncertainty
- Scalability concerns

4. DeFi Major Business Models: Decentralized Payment Services

- Peer-to-peer payments without banks
- Cross-border transfers at lower cost
- Powered by smart contracts
- Integrated with wallets and dApps
- Revenue model: protocol fees



4. DeFi Major Business Models: Decentralized Payment Services

Features

- Cross-border transfers
- Low transaction fees
- Smart contract automation
- Wallet-to-wallet transfers

Examples

- Payment networks built on Ethereum
- Lightning-based systems on Bitcoin

Revenue Model

- Protocol transaction fees
- Validator/miner fees
- Token-based incentives

Benefits

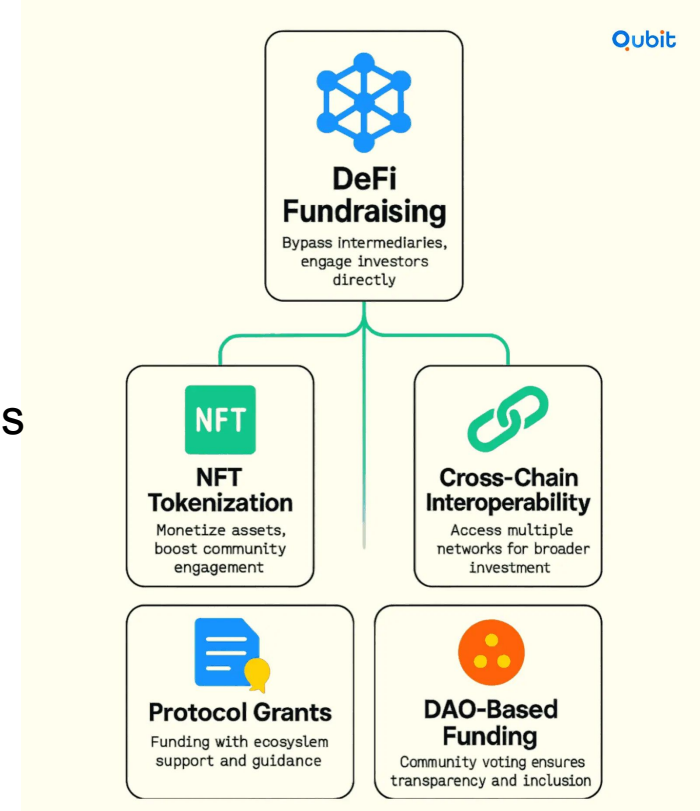
- ✓ Faster settlements
- ✓ Reduced intermediaries
- ✓ Financial inclusion

Risks

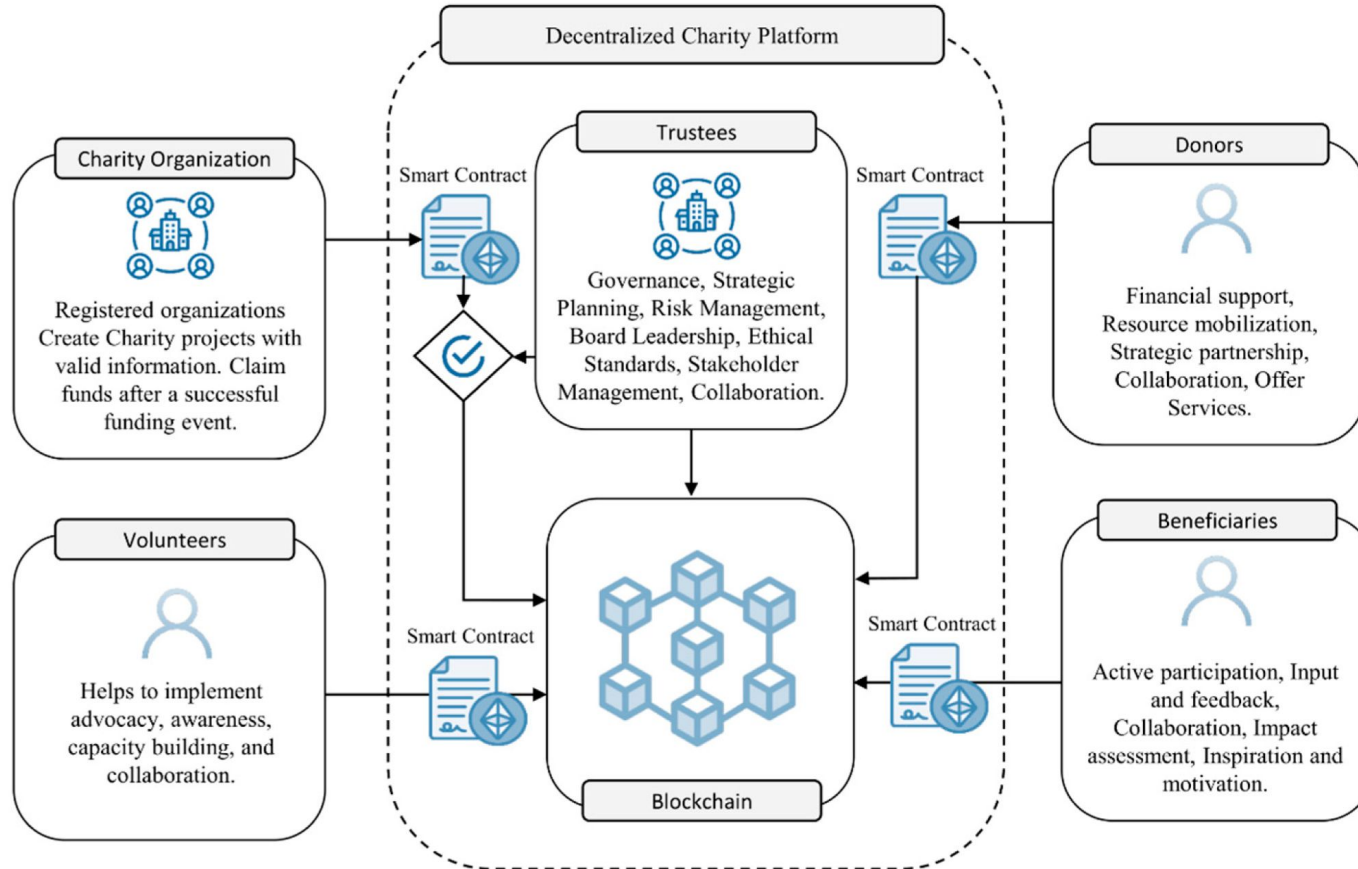
- Network congestion
- Gas fee fluctuations
- Compliance issues

4. DeFi Major Business Models: Decentralized Fundraising

- Capital raising through token issuance
- Includes ICOs, IDOs, and token sales
- Global investor participation
- Transparent fundraising via blockchain
- Revenue model: token appreciation, platform fees



4. DeFi Major Business Models: Decentralized Fundraising



4. DeFi Major Business Models: Decentralized Fundraising

Common Mechanisms

- ICO (Initial Coin Offering)
- IDO (Initial DEX Offering)
- Token sales via smart contracts

How It Works

1. Project issues tokens
2. Investors purchase tokens
3. Funds used for development

Examples : Fundraising platforms using decentralized exchanges like Uniswap

Business Model

- Token appreciation
- Platform listing fees
- Allocation mechanisms

Advantages

- ✓ Global investor participation
- ✓ Transparent transactions
- ✓ Faster capital formation

Challenges

- Regulatory scrutiny
- Investor protection issues
- Scam risks

4. DeFi Major Business Models: Decentralized Fundraising

ICO – Initial Coin Offering : A fundraising method where a project sells its tokens directly to investors without an exchange intermediary.

How It Works

1. Project creates a token.
2. Publishes whitepaper.
3. Investors send crypto (ETH, BTC, etc.).
4. Tokens are distributed to investors.

Key Characteristics

- Fully decentralized
- No exchange involvement
- Open global participation

Famous Example : Ethereum (ICO in 2014)

Advantages

- ✓ Easy to launch
- ✓ Global investor access
- ✓ No exchange approval required

Disadvantages

- High scam risk
- No guaranteed listing
- Limited investor protection

4. DeFi Major Business Models: Decentralized Fundraising

IEO – Initial Exchange Offering: A token sale conducted through a centralized crypto exchange.

How It Works

1. Project partners with exchange.
2. Exchange vets the project.
3. Token sale happens on exchange platform.
4. Tokens are listed after sale.

Key Characteristics

- Exchange acts as intermediary
- Mandatory KYC
- Immediate exchange listing

Example Platforms

- Binance Launchpad
- KuCoin Spotlight

Advantages

- ✓ Higher investor trust
- ✓ Built-in liquidity
- ✓ Exchange marketing support

Disadvantages

- High listing fees
- Centralized control
- Restricted country participation

4. DeFi Major Business Models: Decentralized Fundraising

IDO – Initial DEX Offering: A token sale conducted on a decentralized exchange (DEX).

How It Works

1. Project launches token on DEX.
2. Liquidity pool is created.
3. Investors buy via decentralized trading.

Key Characteristics

- No centralized exchange
- Smart contract-based
- Instant liquidity

Example DEX Platforms

- Uniswap
- PancakeSwap

Advantages

- ✓ Immediate trading
- ✓ Lower listing costs
- ✓ More decentralized

Disadvantages

- High volatility
- Bot manipulation risk
- Less project screening

4. DeFi Major Business Models: Decentralized Fundraising

Feature	ICO	IEO	IDO
Platform	Project website	Centralized Exchange	Decentralized Exchange
Intermediary	No	Yes (CEX)	No (Smart Contract)
KYC	Usually No	Yes	Usually No
Liquidity	Not guaranteed	Immediate	Immediate
Risk Level	High	Medium	Medium-High
Regulation	Minimal	More regulated	Limited regulation

4. DeFi Major Business Models: Decentralized Contracting

- Smart contracts automate agreements
- Self-executing and trustless
- Used in lending, insurance, derivatives
- Reduces need for intermediaries
- Revenue model: execution fees

4. DeFi Major Business Models: Decentralized Contracting

Applications

- Decentralized lending
- Insurance
- Derivatives
- Asset tokenization

Revenue Model

- Execution fees
- Protocol governance tokens
- Collateral-based interest spreads

Advantages

- ✓ Automation
- ✓ Reduced fraud
- ✓ Cost efficiency

Risks

- Smart contract bugs
- Code vulnerabilities
- Oracle dependency risks

4. DeFi Major Business Models : Comparison

Model	Core Function	Revenue Source	Risk Level
Decentralized Currencies	Store of value & exchange	Transaction fees	Medium
Payment Services	Transaction processing	Network fees	Medium
Fundraising	Capital formation	Token sales	High
Contracting	Automated agreements	Execution fees	Medium-High